Lecture 18

EE 421 / CS 425

Digital System Design

Fall 2026

Shahid Masud

Topics

Quiz Next Week

- Floating Point Adder and Multiplier Operation (quick recap)
- Some Examples of Circuit Design for Floating Point Module
- ------- ------------- -------
- Introduction to **Programmable Logic Hardware Device**
- Nomenclature and types
- Logic Array Circuits
- PLA and PAL
- Programmable Logic Devices – PLD

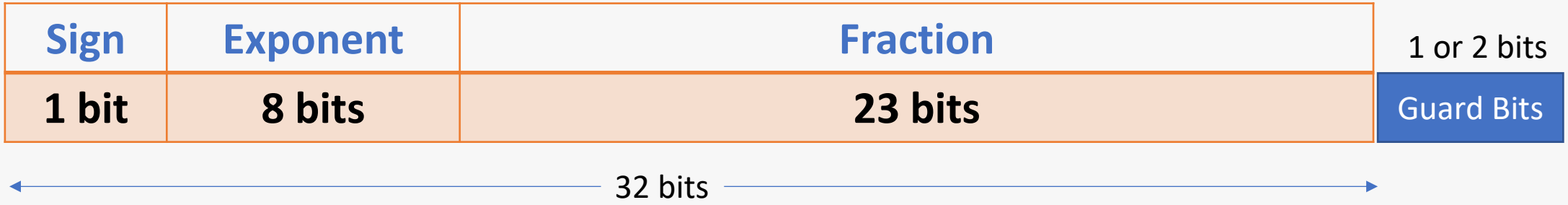
Floating Point Arithmetic – Digital Design

$N=(-1)^S \times (1+F) \times 2^E$ E.g. $91.820734 \times 10^{-34}$

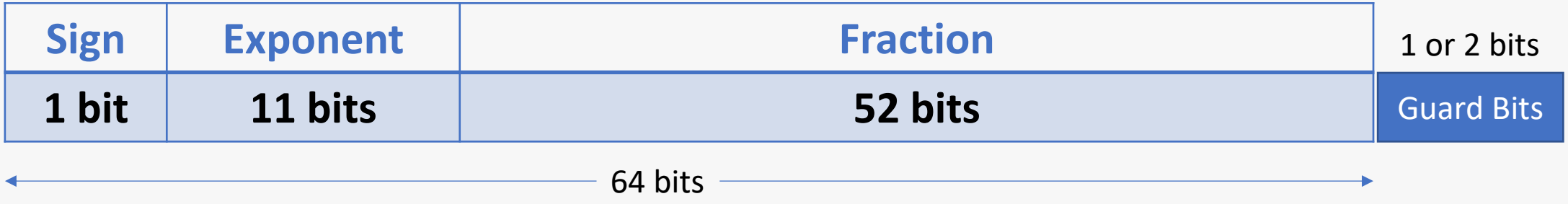
- A signed-magnitude system for the fractional part and a biased notation for the exponent
- Three subfields
  - Sign S
  - Fraction F (or Significand or Mantissa)
  - Exponent E
- Sign bit is 0 for positive numbers, 1 for negative numbers
- Fractions always start from **1**.xxxx, hence the integer **1** is not written (register has xxxx)
- Exponent is biased by +127 (add 127 to whatever is in register bits)
- Normalize: Express numbers is the standard format by shifting of bits and adding / subtracting from Exponent register

IEEE 754 Floating Point Representation

Single Precision IEEE 754



Double Precision IEEE 754



Examples of Floating Point Representation

$-(13.45)_{10}$
$= (1101.01\ 1100\ 1100\ 1100\ \ldots\ldots)^2$; this is un-normalized
$= (1.10101\ 1100\ 1100\ 1100\ 1100\ 1) \times 2^3$; normalized
Fraction part is 10101 1100 1100 1100 1
Biased Exponent is 3+127 = 130
Sign = 1

5.0345
= 101 . 0000 1000 1101 0100 1111 110; this is un-normalized
$= 1.\ 01\ 0000\ 1000\ 1101\ 0100\ 1111\ 110 \times 2^2$; normalized
Biased Exponent = 2+127 = 129 = $(1000\ 0001)_2$
Fraction = 01 0000 1000 1101 0100 1111 110
Sign = 0

3 Distinct Steps:
Evaluate Normalization
Biased Exponent
Fraction or Mantissa

Floating Point Arithmetic – Digital Design

$N=(-1)^S \times (1+F) \times 2^E$ E.g. $91.820734 \times 10^{-34}$

- A signed-magnitude system for the fractional part and a biased notation for the exponent
- Three subfields
  - Sign S
  - Fraction F (or Significand or Mantissa)
  - Exponent E
- Sign bit is 0 for positive numbers, 1 for negative numbers
- Fractions always start from **1**.xxxx, hence the integer **1** is not written (register has xxxx)
- Exponent is biased by +127 (add 127 to whatever is in register bits)
- Normalize: Express numbers is the standard format by shifting of bits and adding / subtracting from Exponent register

Floating Point Multiplication

Consider two floating point numbers:
$(F_1 \times 2^{E1})$ and $(F_2 \times 2^{E2})$

Beginning Denormalize

The product of these two numbers is:

$= (F_1 \times 2^{E1}) \times (F_2 \times 2^{E2})$

$= (F_1 \times F_2) \times 2^{(E1+E2)}$

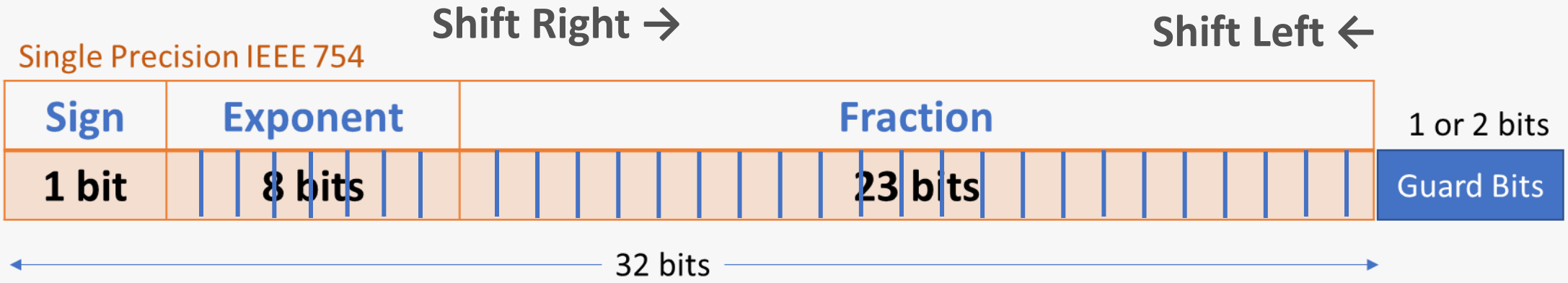
$= F \times 2^{E}$

Evaluate Fraction Multiplication
Evaluate Biased Exponent

Sign of result depends on Sign of the two numbers

Final Normalize

Complexity of Normalize and DeNormalize



What is the Complexity, in terms of time and area?
Can this be improved?

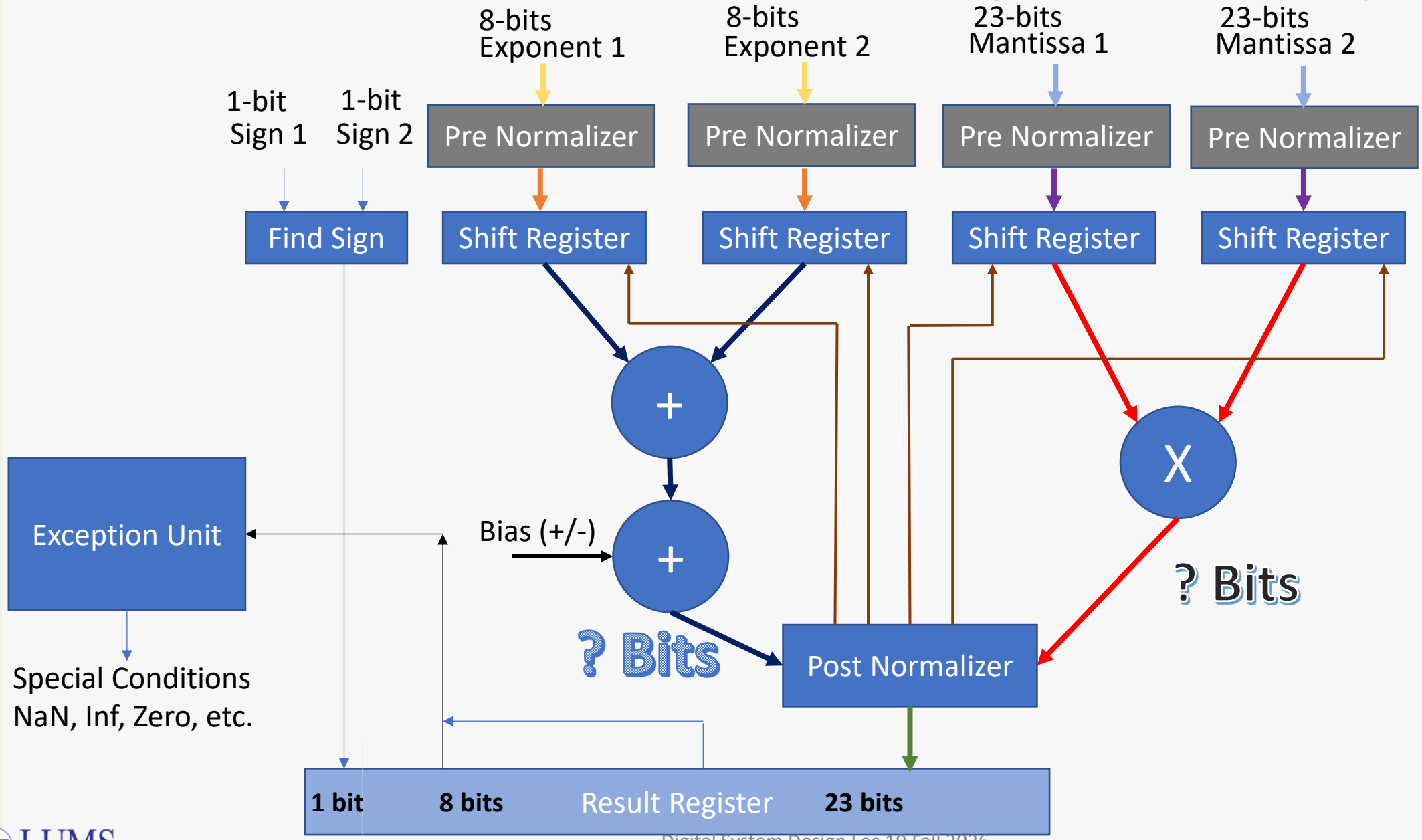
Floating Point Addition

Consider two floating point numbers:
$(F_1 \times 2^{E1})$ and $(F_2 \times 2^{E2})$

Initial Denormalize
The addition of these two numbers is:
$= (F_1 + F2) \times (2^{Enew})$
Enew is obtained by Shl or Shr of F1 and F2 so that E1 becomes same as E2
$= F \times 2^{Enew}$
Final Normalize
Take into account Guard Bits

Sign of result depends on Sign of the two numbers

Data Path Architecture Floating Point Multiplier



Look at Some Examples of FP Design

Use Novelty in Multipliers:

Booth or Radix-4 Encoding

Approximate Computations

Use Compressed Adders for Adder Chains

Programmable Logic and Reconfigurable Logic

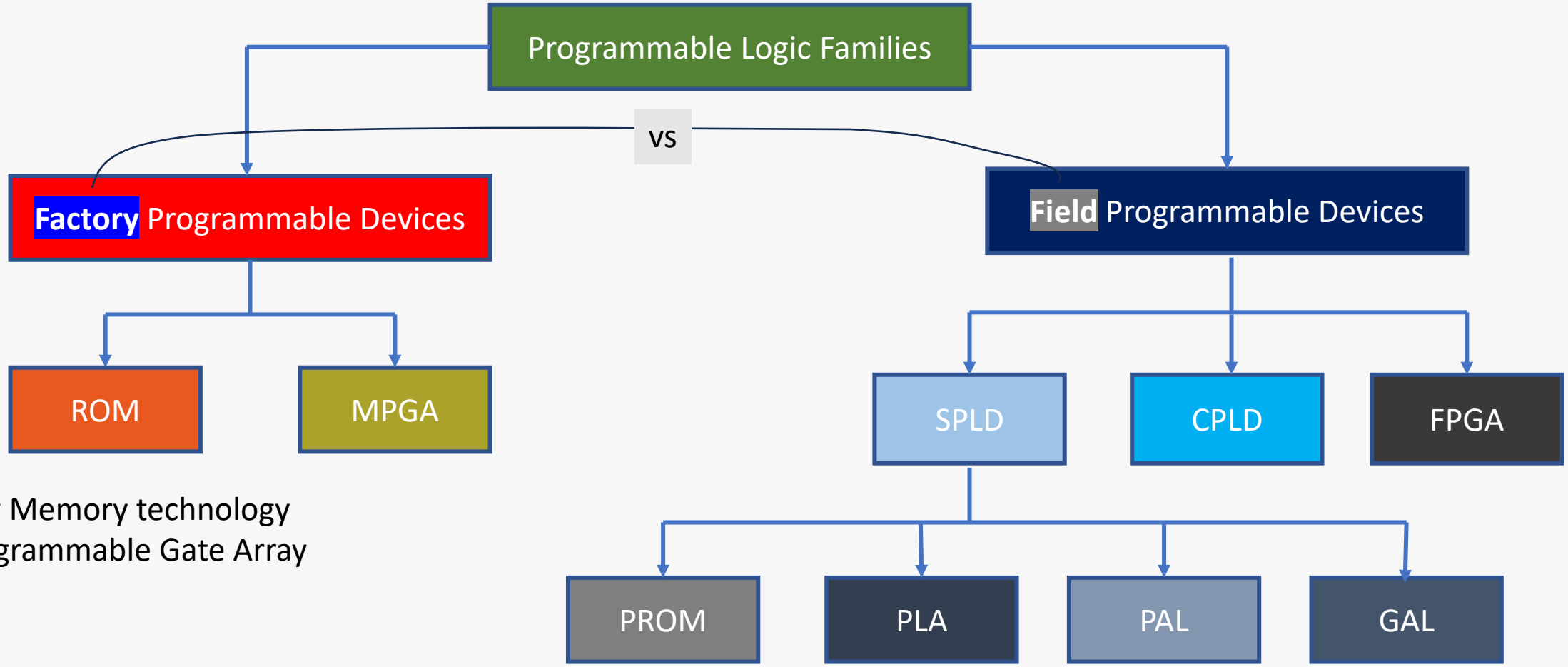
Definition of Programmable Logic

- A family of semiconductor integrated circuits that can be reconfigured through:
  - special hardware-description-languages, and
  - associate Electronic Design Automation tools
  - Special configuration equipment

to produce a variety of hardware functionality from the same integrated circuit chip.

- The functionality in hardware can be erased and reconfigured in a manner analogous to design of a software system using microprocessor.
- All other types of semiconductors have fixed and pre-defined functionality.
- Performance vs Flexibility tradeoffs exist, as with any other digital system.

Programmable Logic Broad Classification

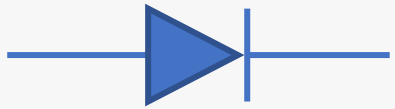


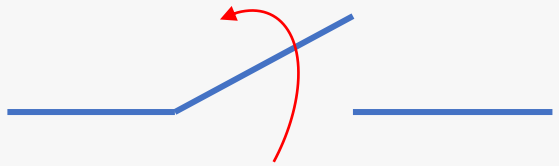
Read only Memory technology
Mask Programmable Gate Array

Simple and Complex Programmable Logic Devices
Field Programmable Gate Array
Generic Array Logic (complex PAL by lattice semi), Programmable Logic Array

Diode – Principle of operation



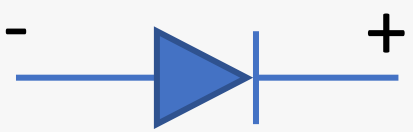






Current flow

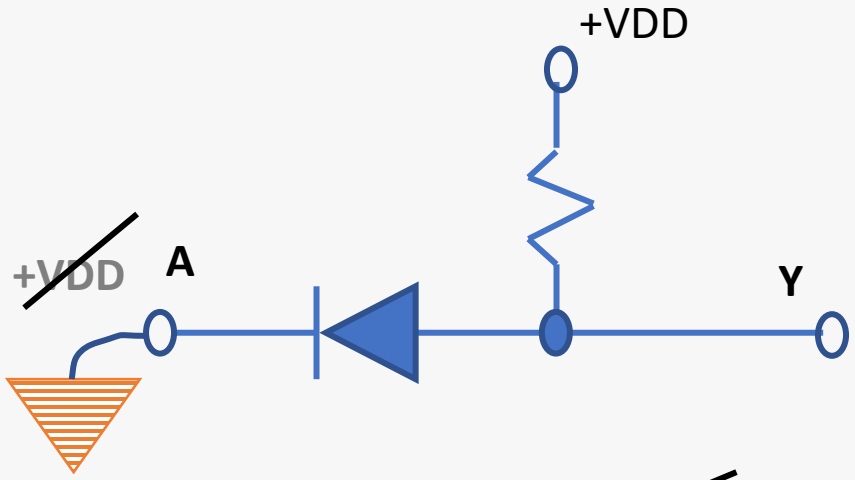
Forward Biased



X Current flow

Reverse Biased

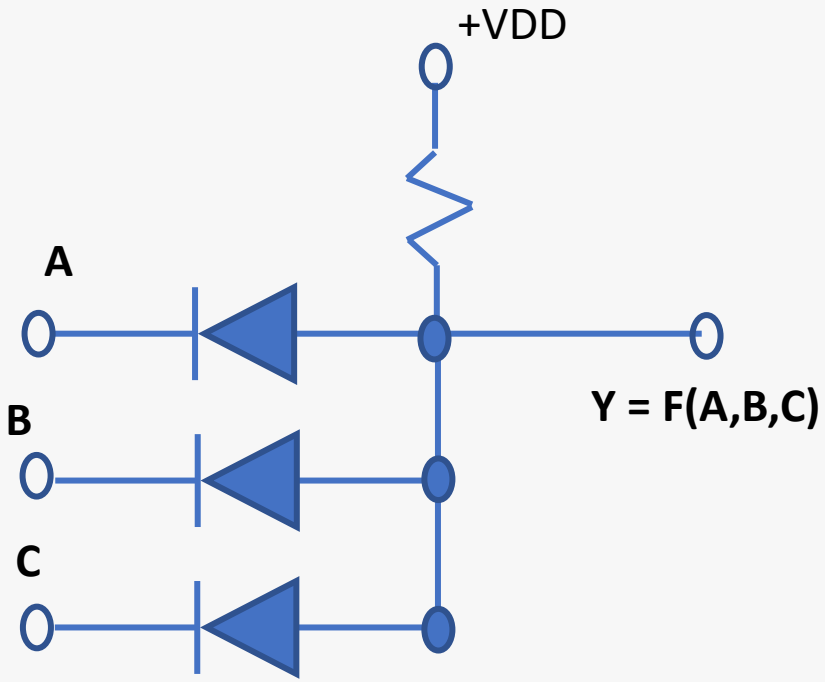
Diode in logic gates



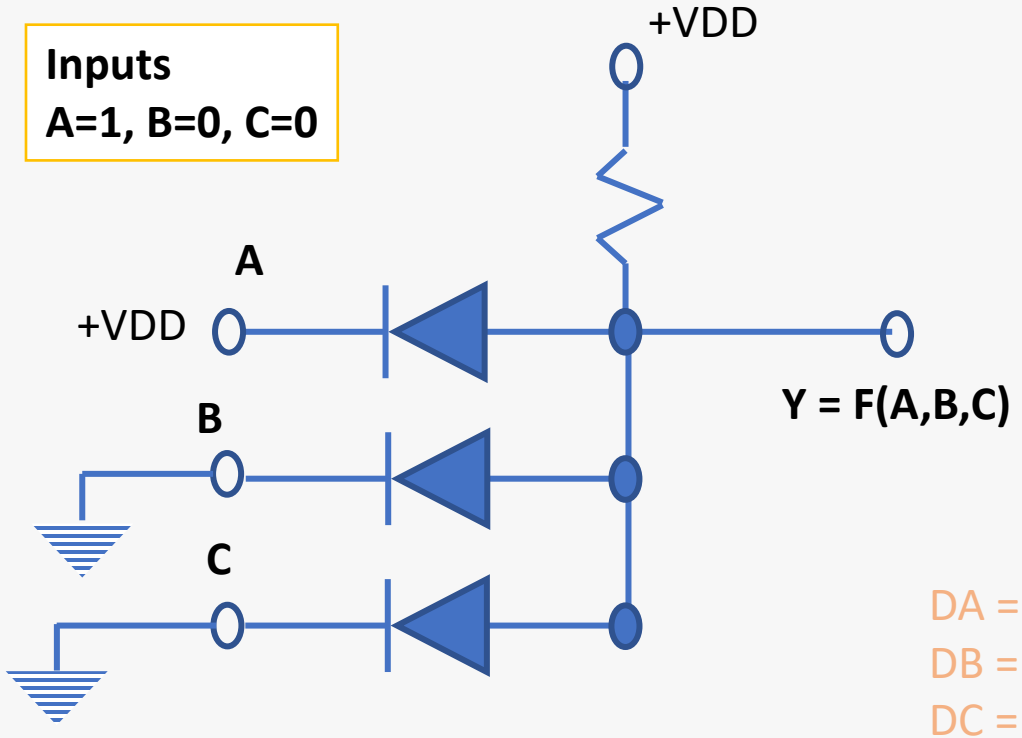
When A=1, Diode is Reverse Biased
Hence Output Y = 1 through +VDD

When A=0, Diode is Forward Biased
Hence Output Y = 0 through A = 0 Volt

Diode based AND Logic Array



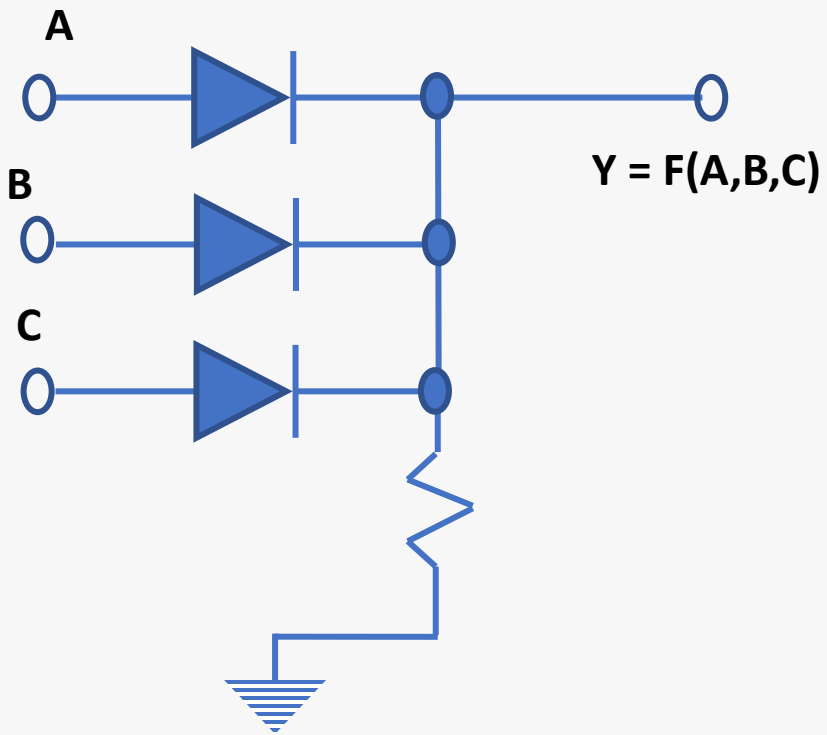
Inputs
A=1, B=0, C=0



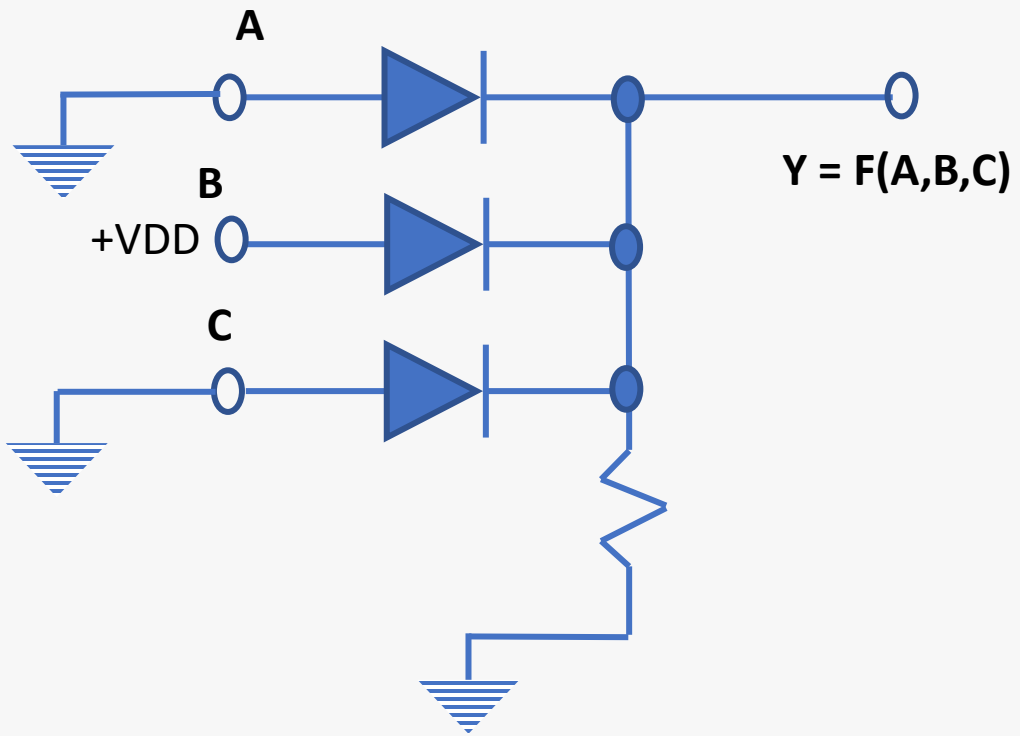
DA = Diode A
DB = Diode B
DC = Diode C

DA=Open Circuit, DB = Forward Biased Short, DC = Forward Biased Short
Y= Logic '0' as it is Shorted to Zero due to DB or DC

Diode based OR array

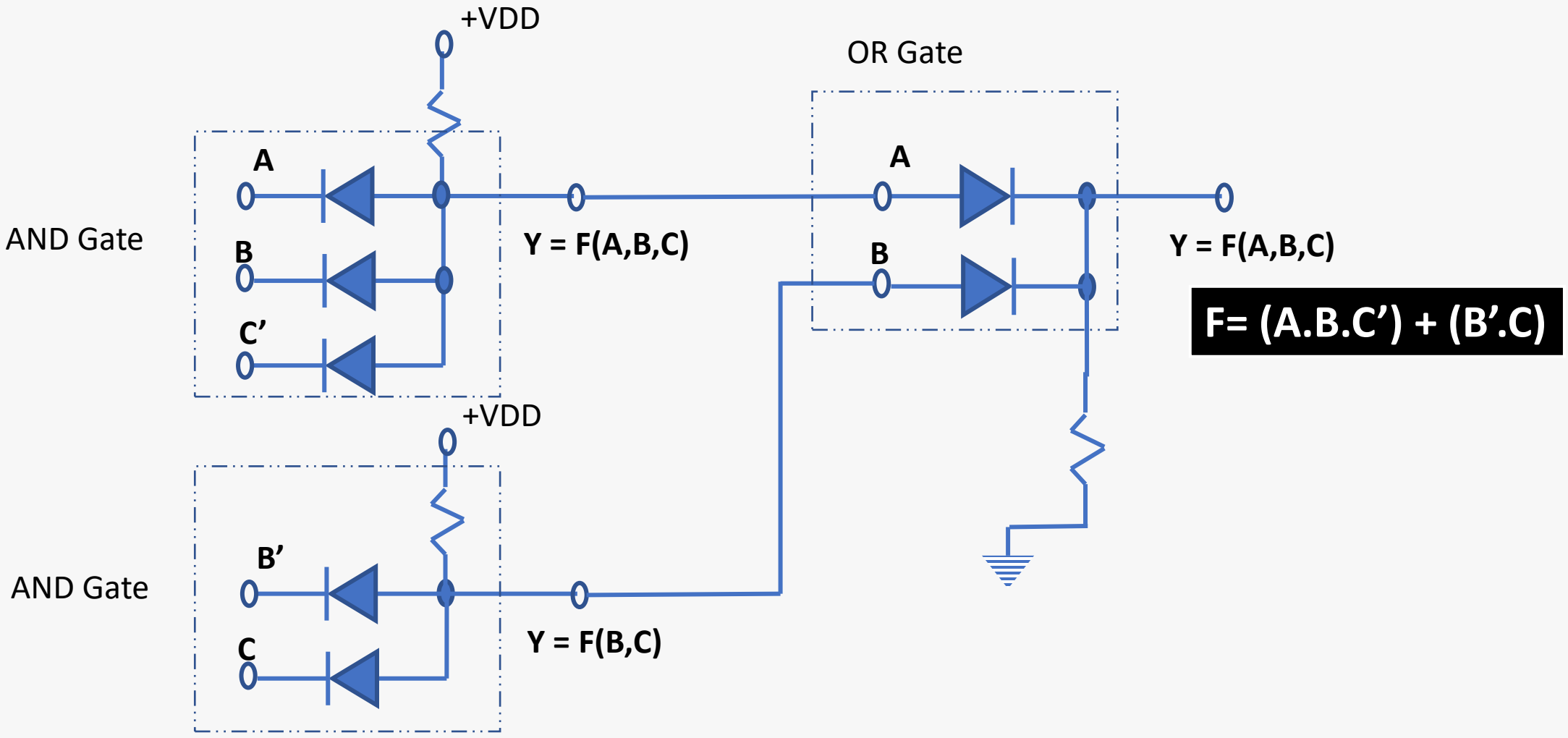


Inputs A = 0, B = 1, C = 0



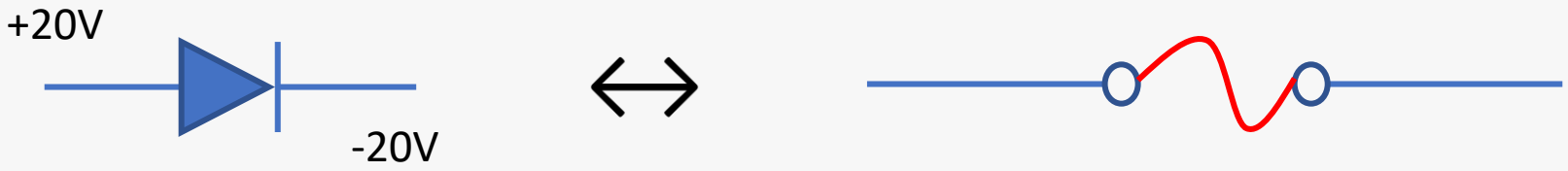
DA and DC = Open Circuit due to Zero input, DB = Short Circuit due to +VDD input
Y = +VDD Logic '1' due to DB being short circuit, other paths are open circuit

Diode based AND-OR Array



Diode as fuse

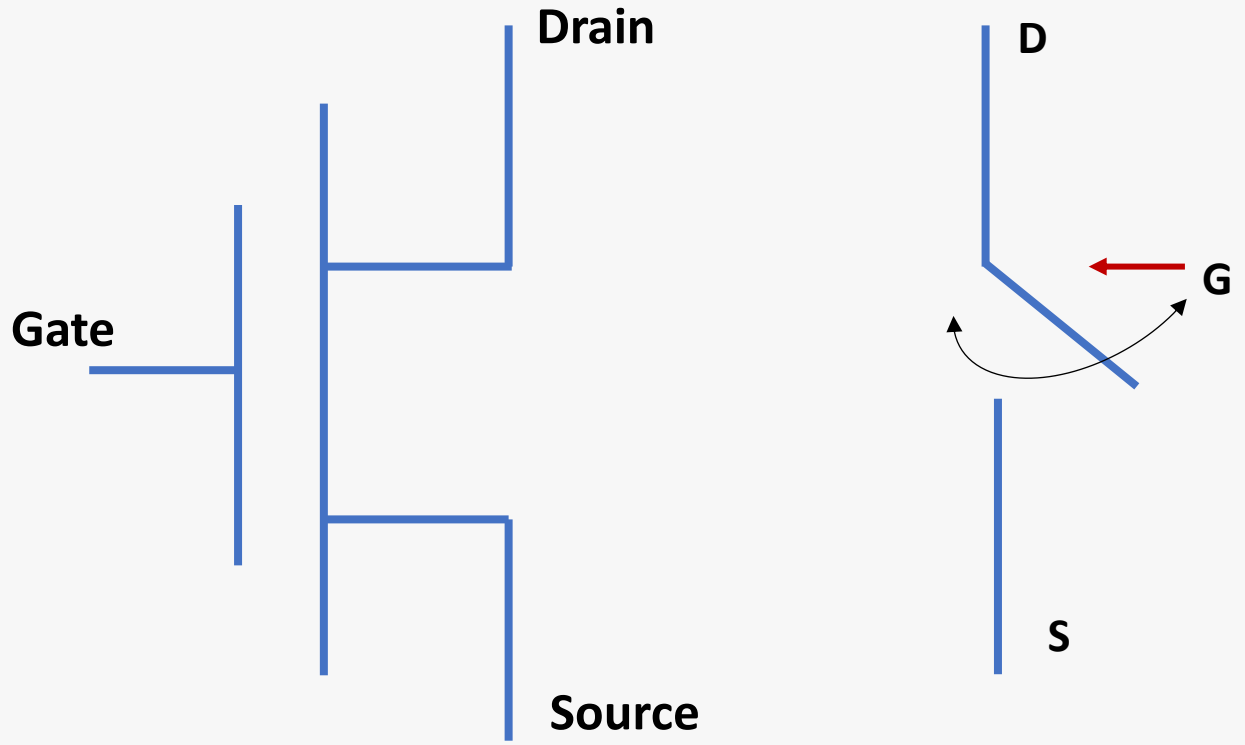
Momentarily apply a higher voltage where open circuit path is required



Desired paths from AND gates and OR gates will be retained, others will be blown out

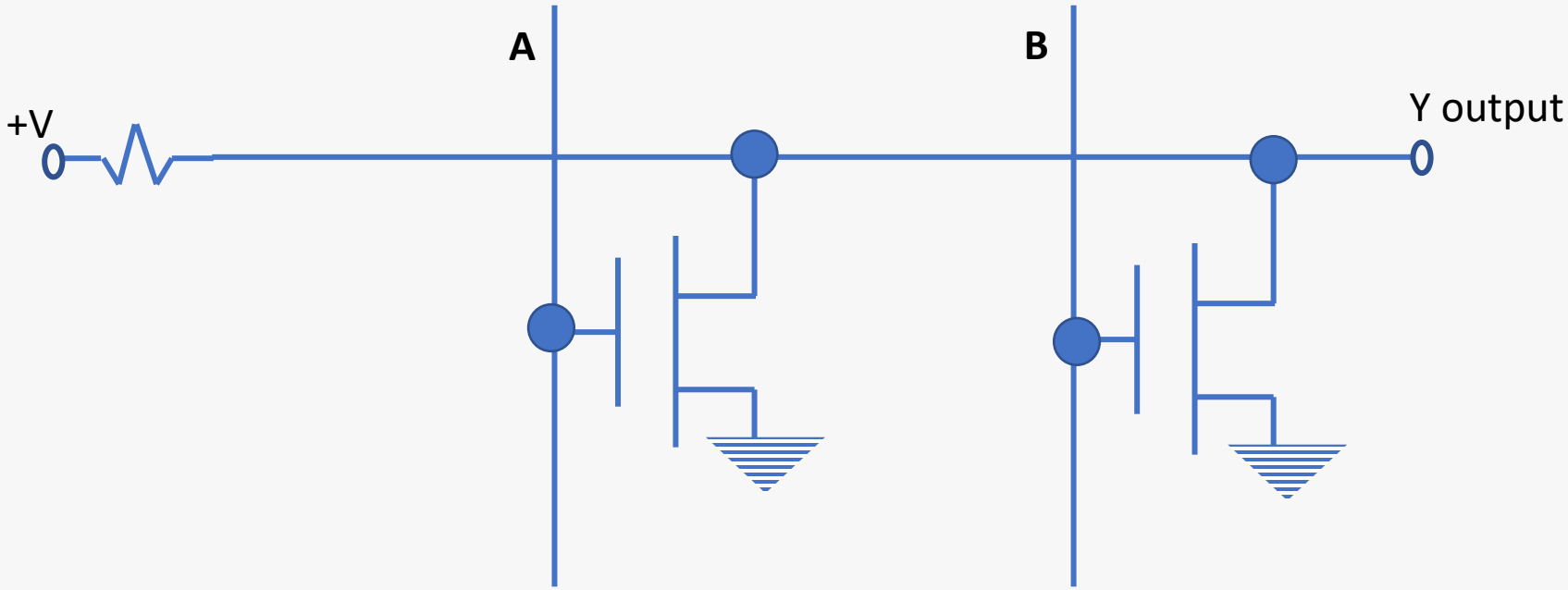
Desired Paths are selected through a MASK during final stages of FABRICATION PROCESS

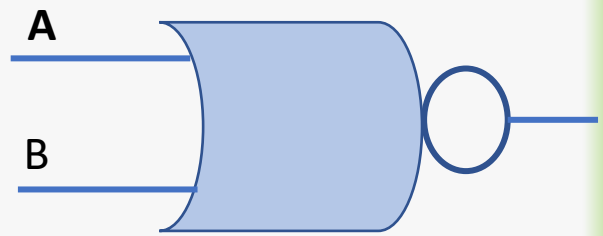
MOS Transistor as Switch



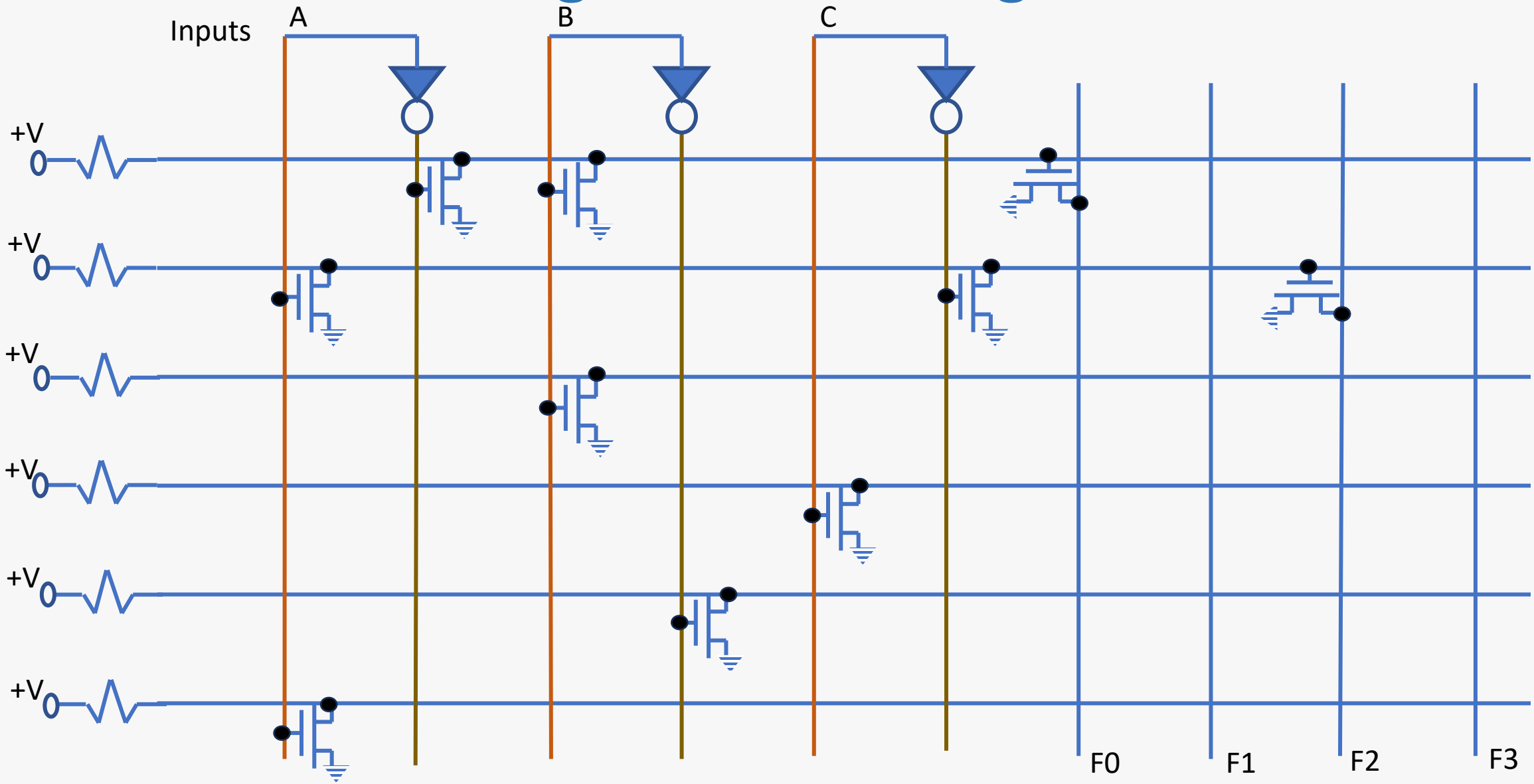
Programming is achieved through controlling GATE Voltage
High GATE Voltage will ensure flow of current, like a short circuit
Low GATE Voltage will ensure no flow of current, like an open circuit

Logic Gates Design with Transistor Switches





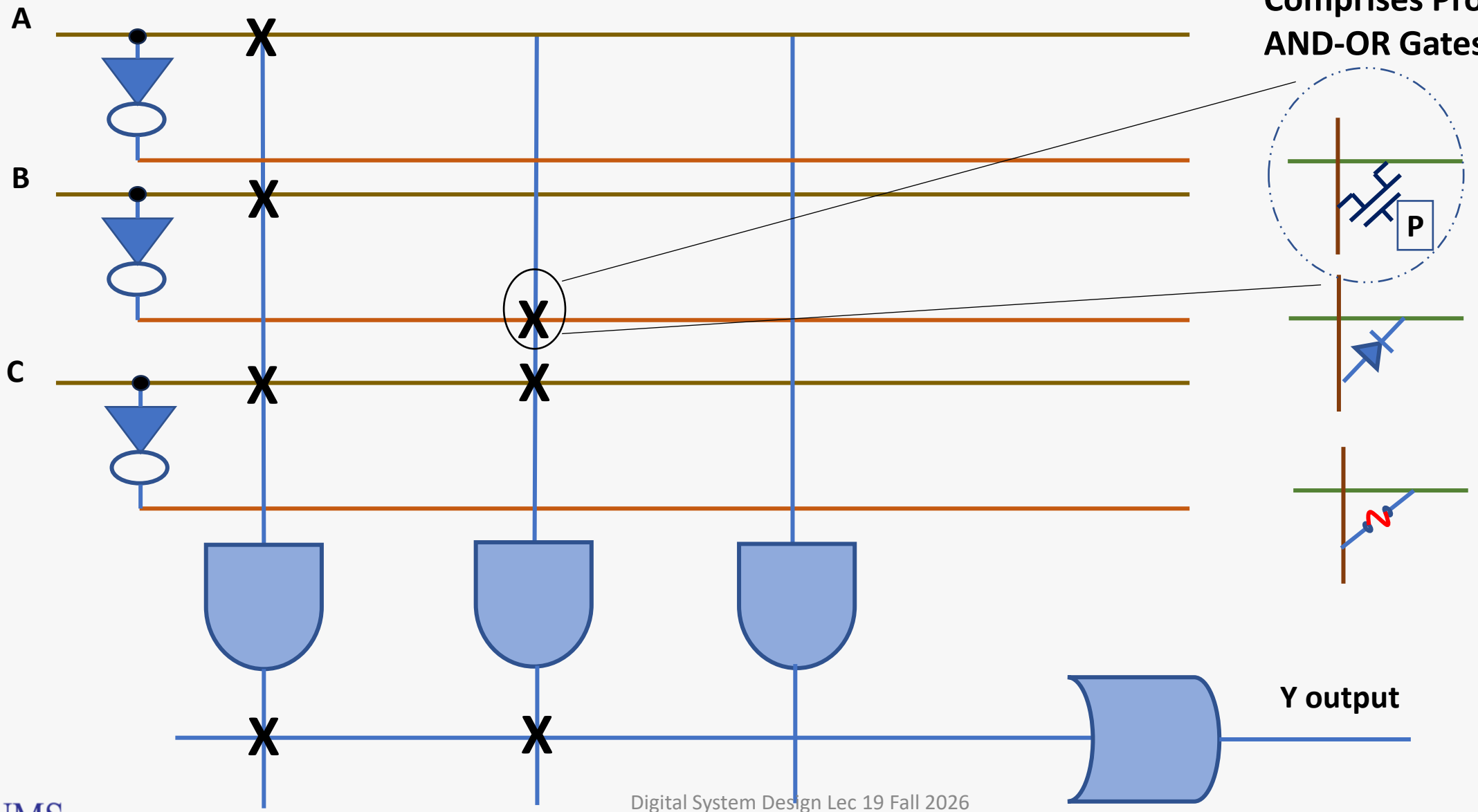
Transistors as Programmable Logic Switches



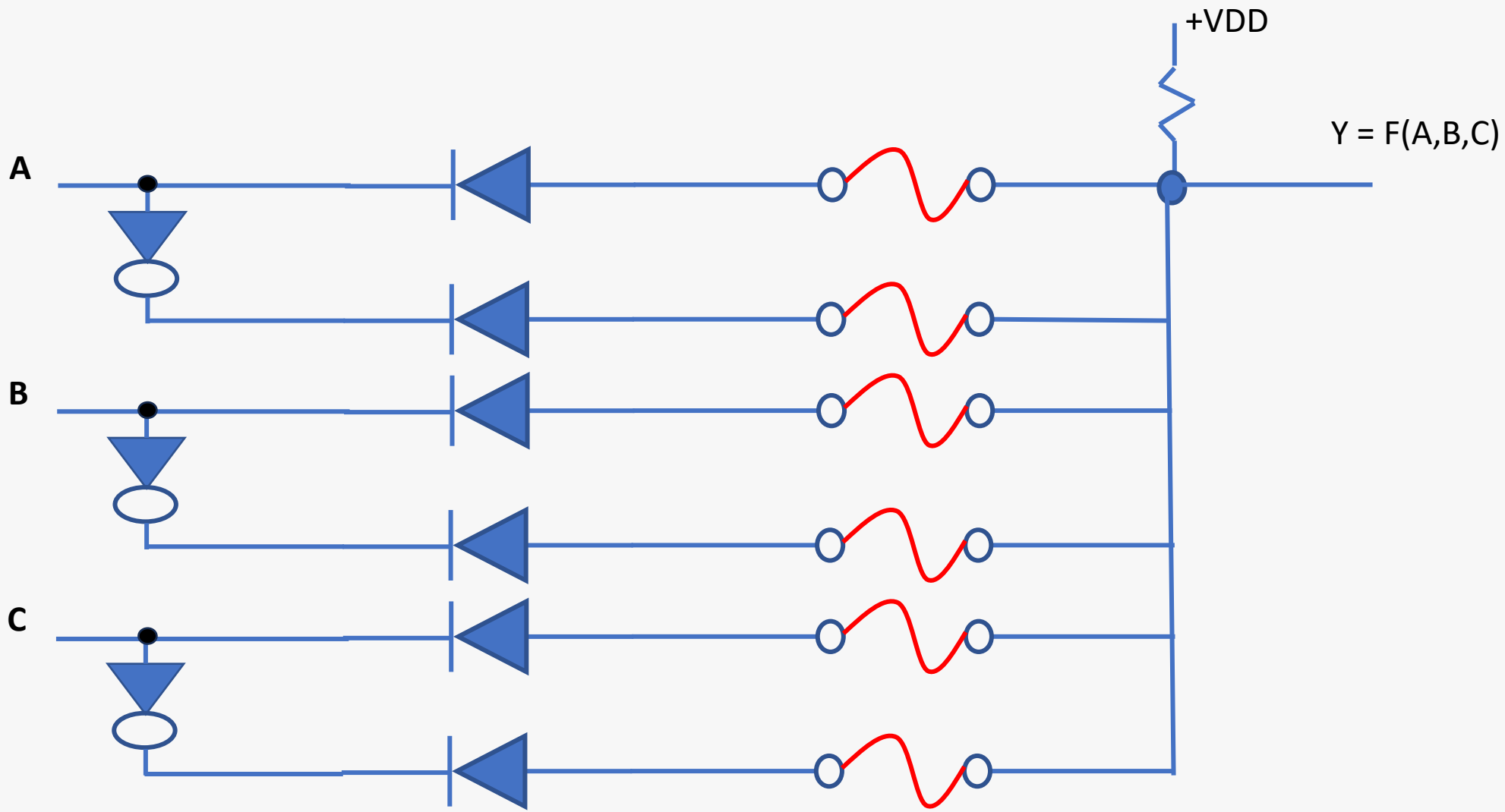
Different logic functions are realized through transistor switches

AND-OR Logic Array

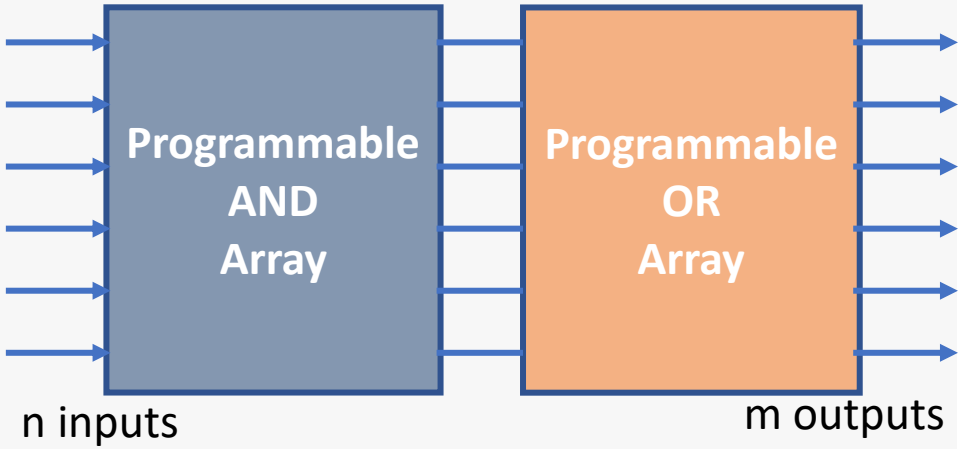
PLA = Programmable Logic Array Comprises Programmable AND-OR Gates

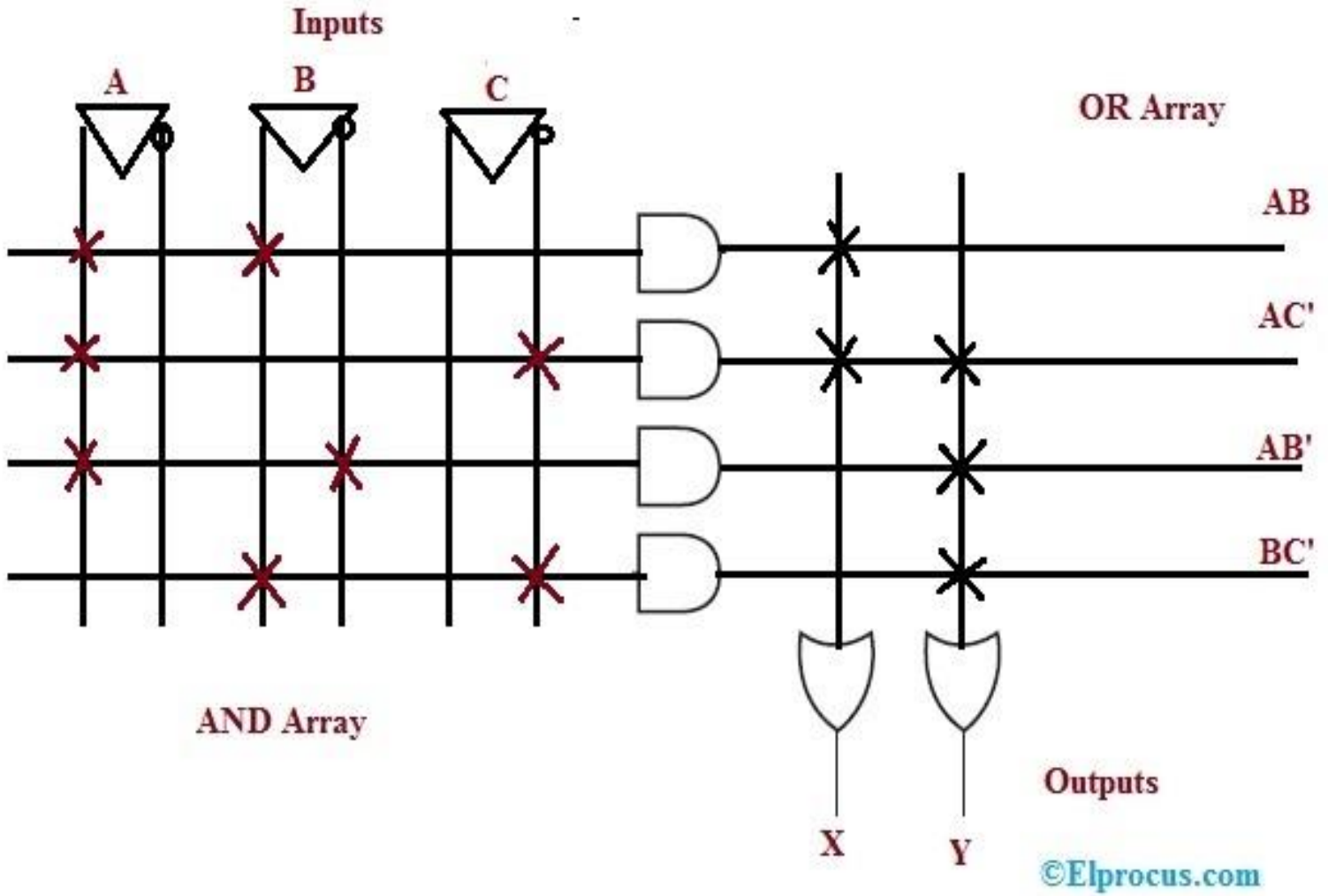


Fuse Programmable Logic Array



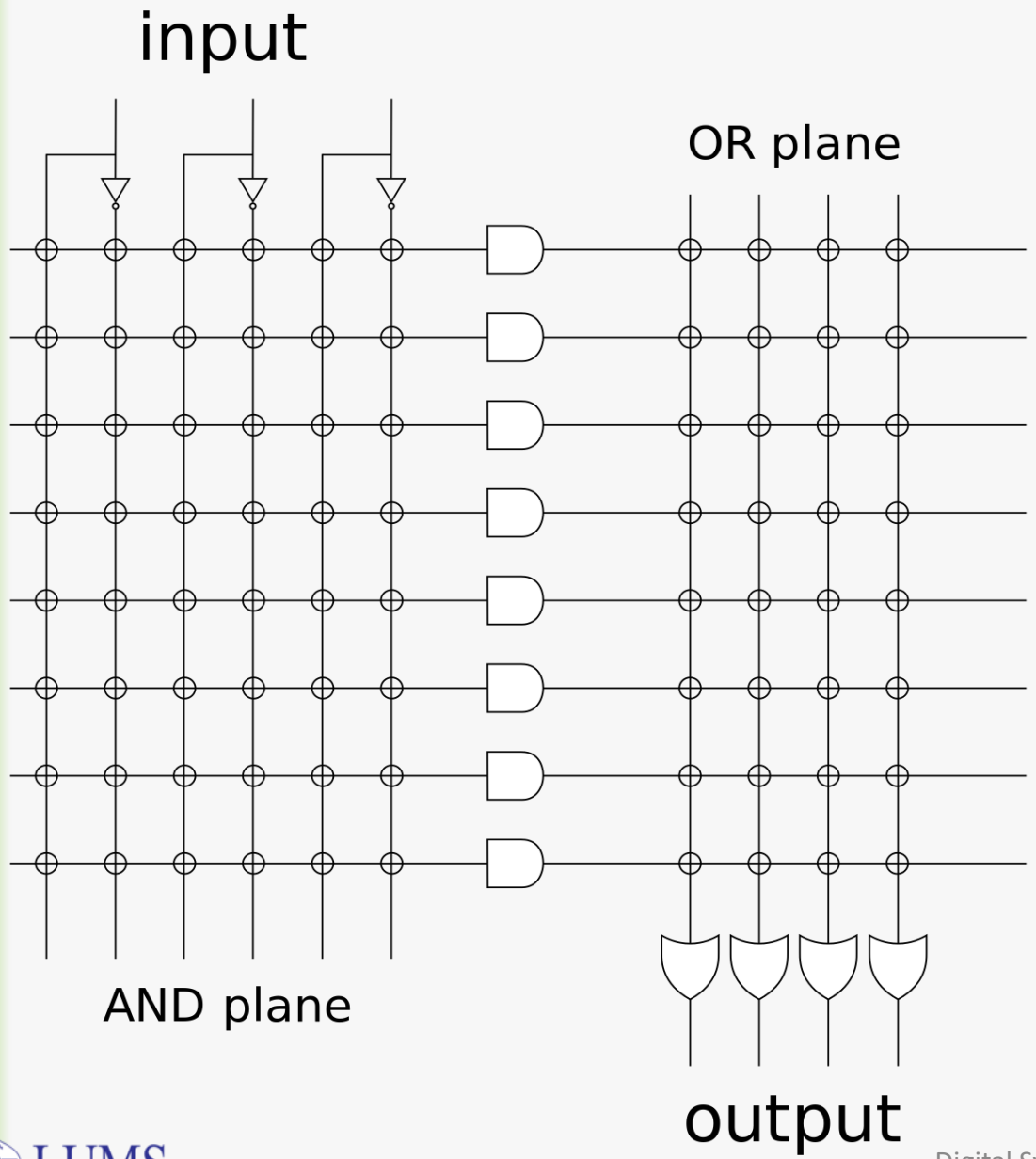
PLA – Programmable Logic Array

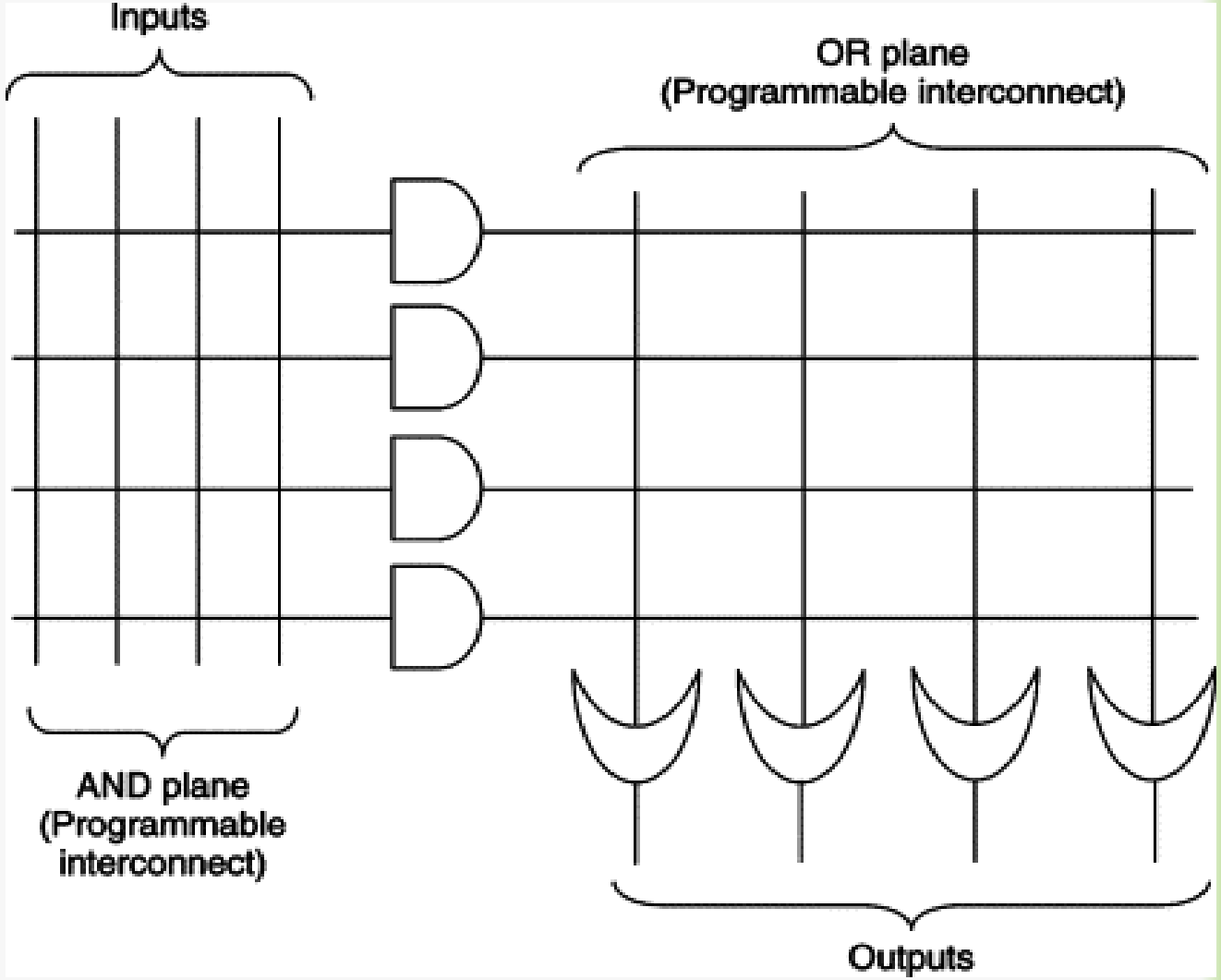




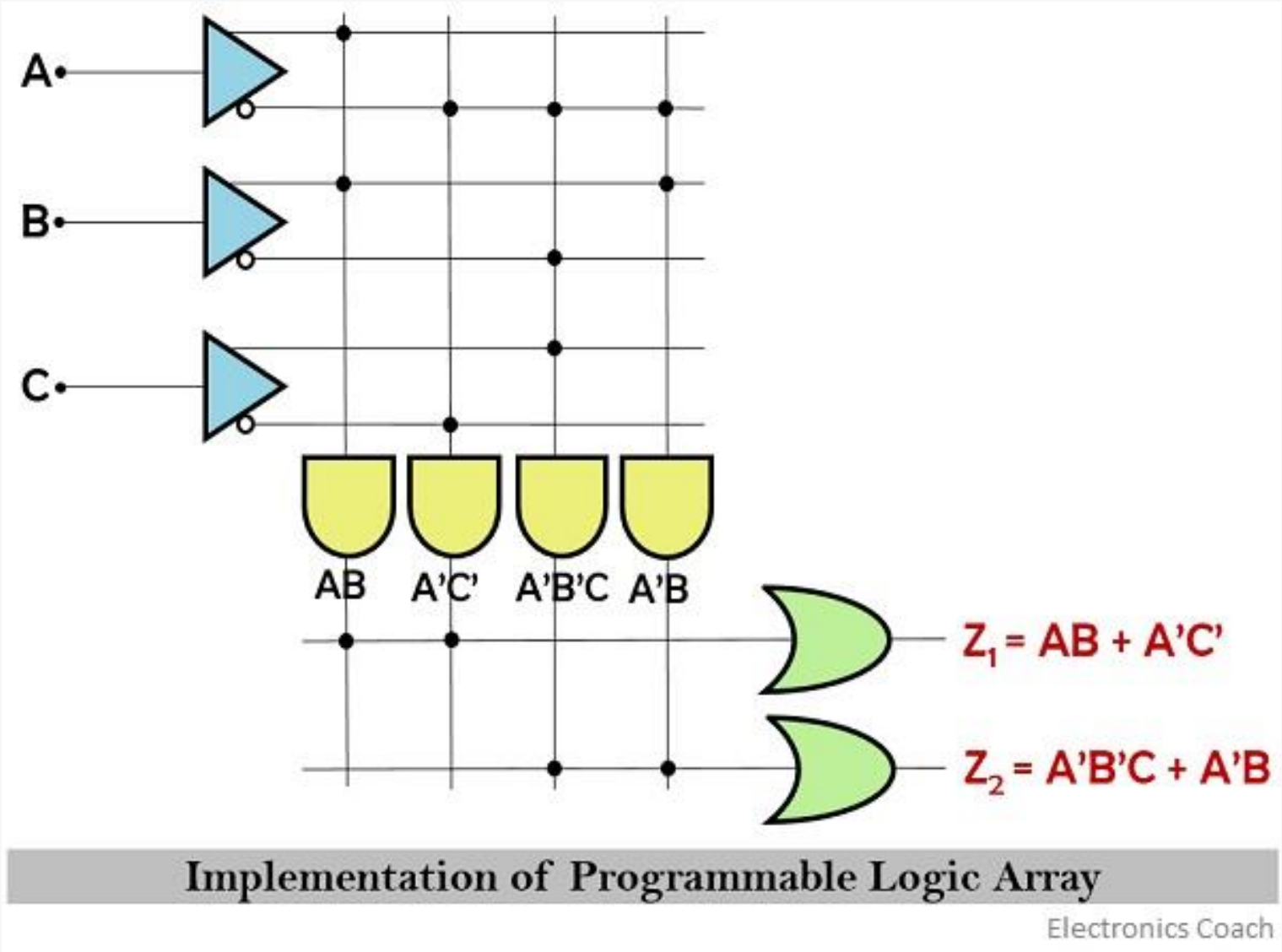
https://www.elprocus.com/what-are-pal-and-pla-design-and-differences/

For Exercise





Example of PLA

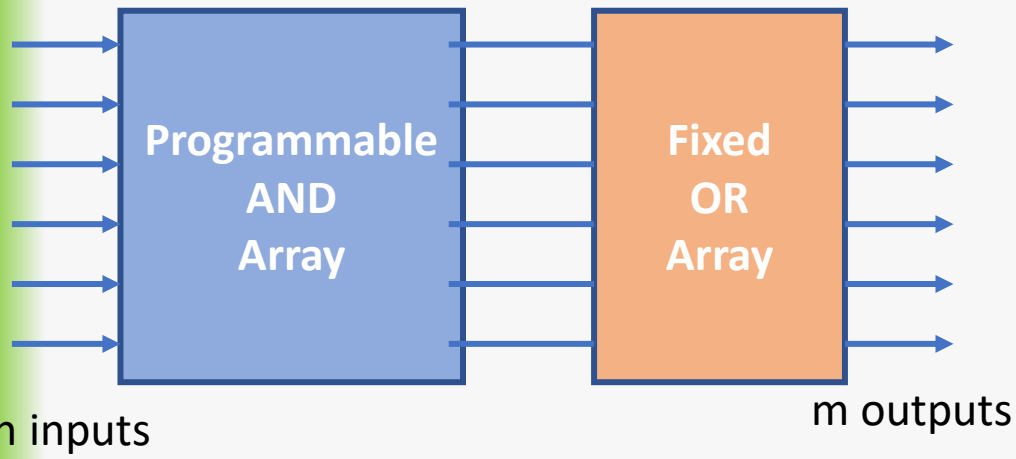


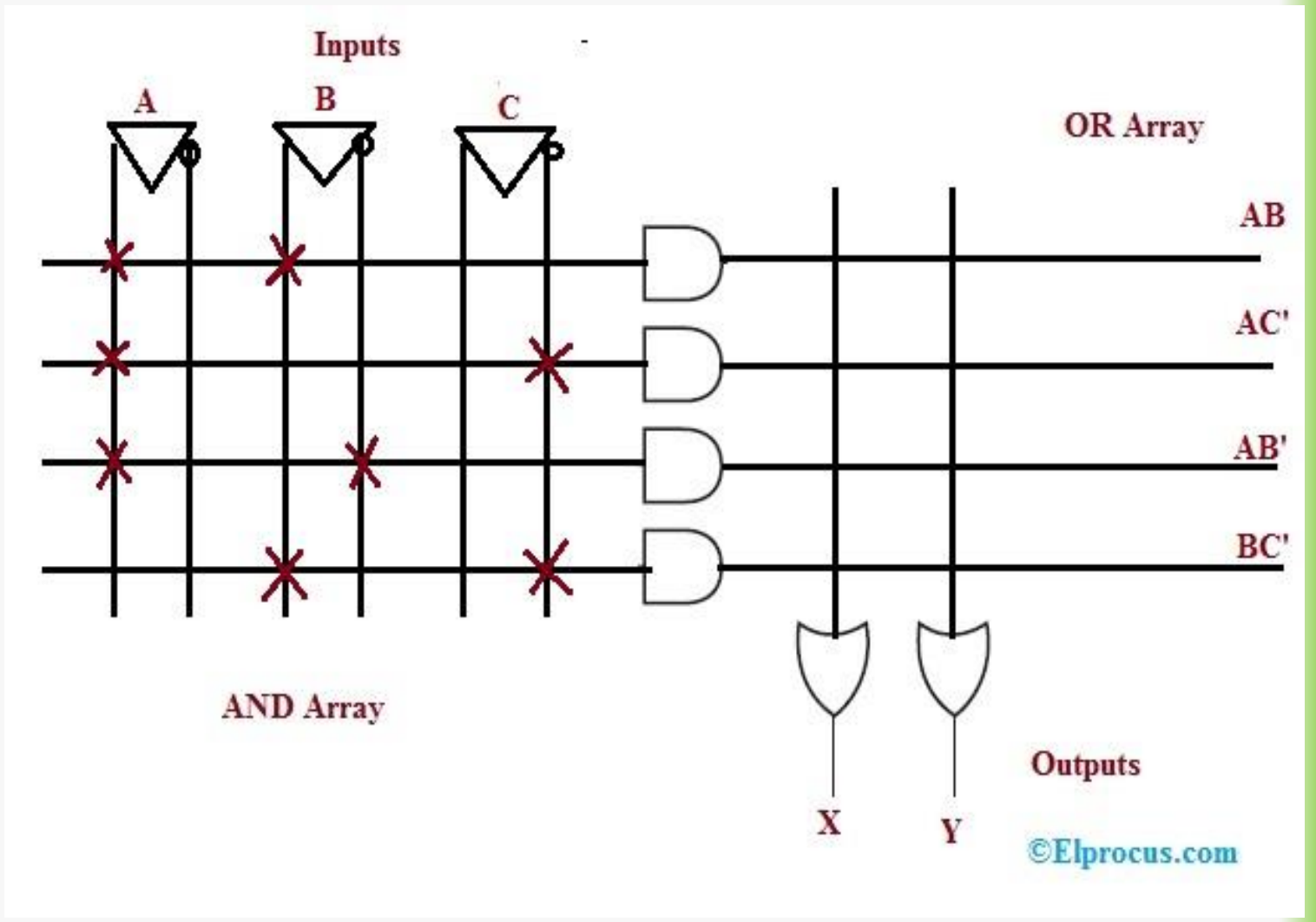
Implementation of Programmable Logic Array

Electronics Coach

https://electronicscoach.com/programmable-logic-array.html

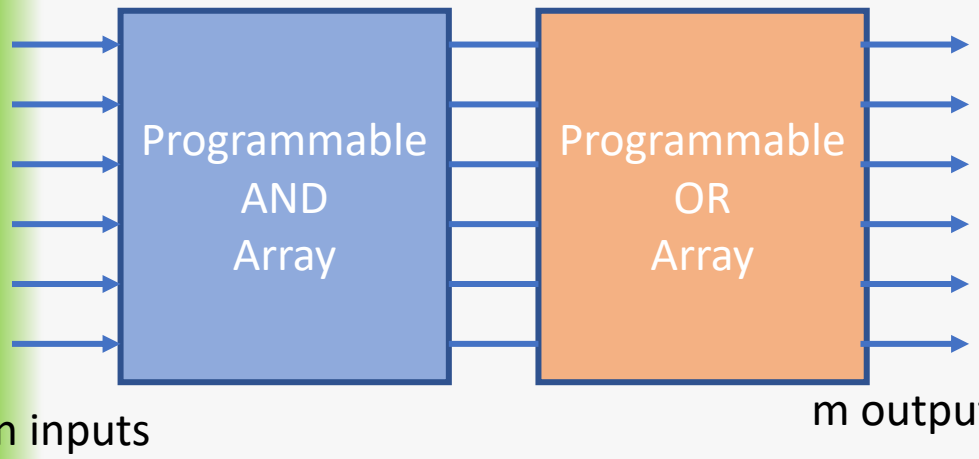
PAL – Programmable Array Logic

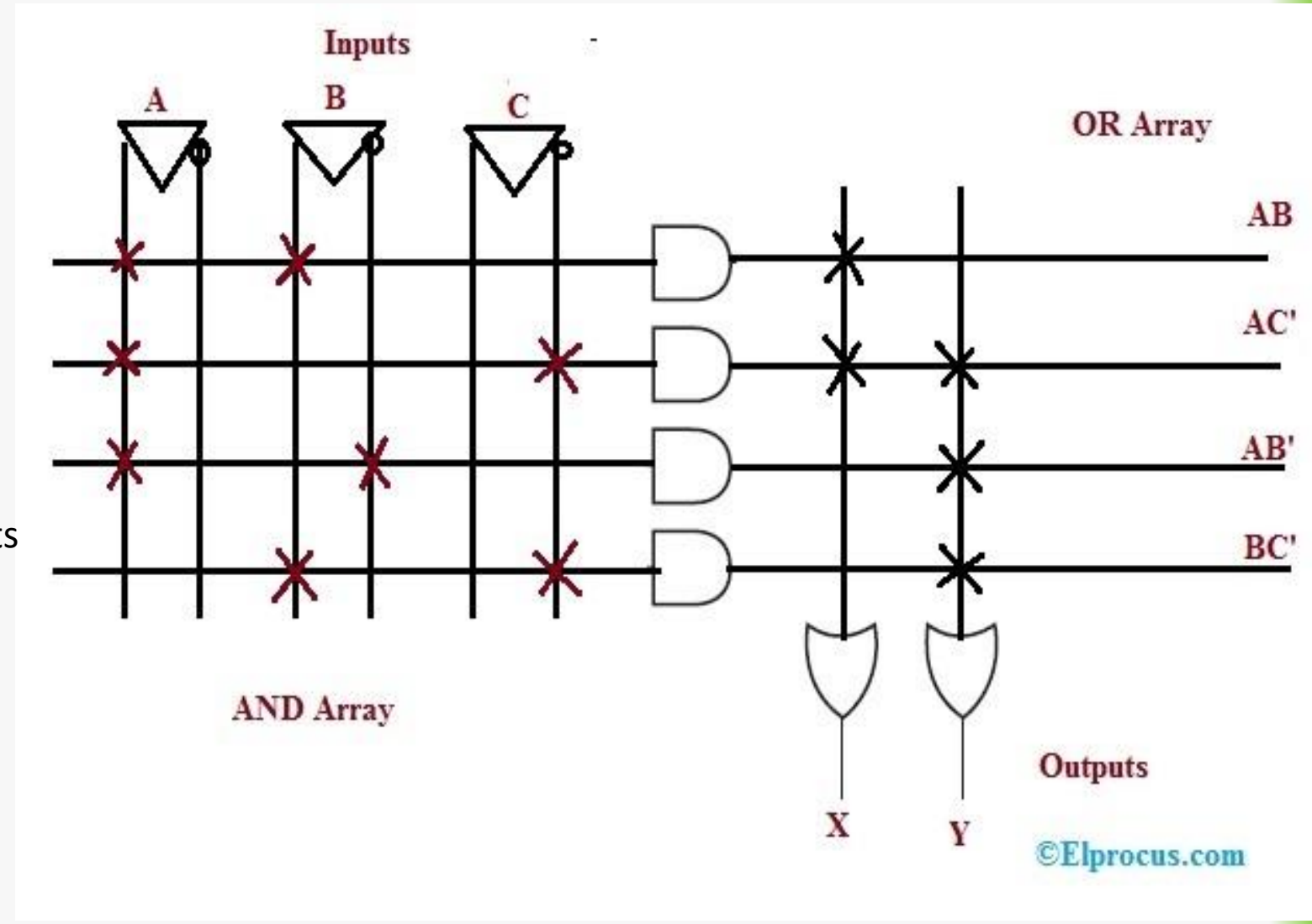




https://www.elprocus.com/what-are-pal-and-pla-design-and-differences/

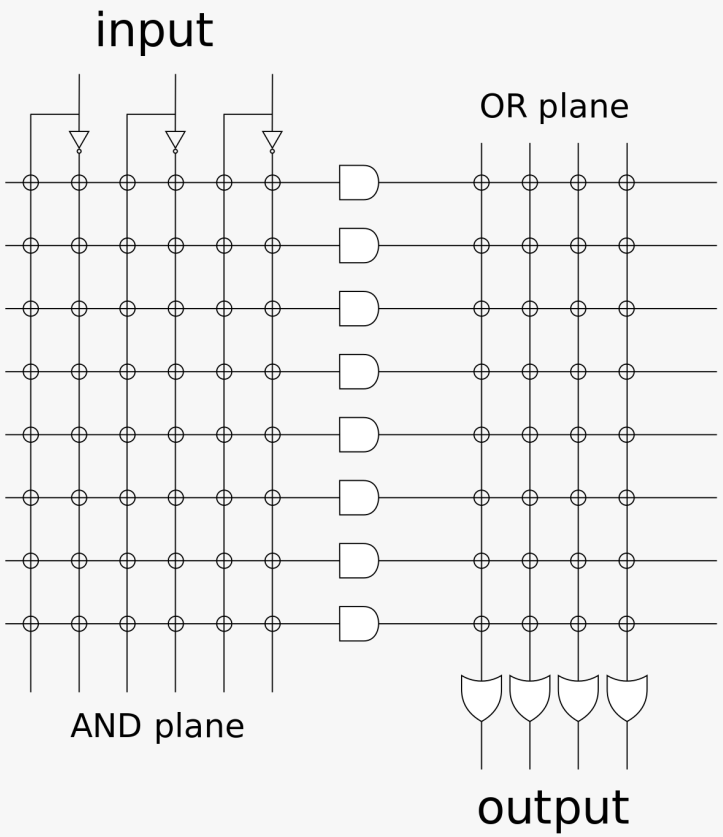
PLA – Programmable Logic Array

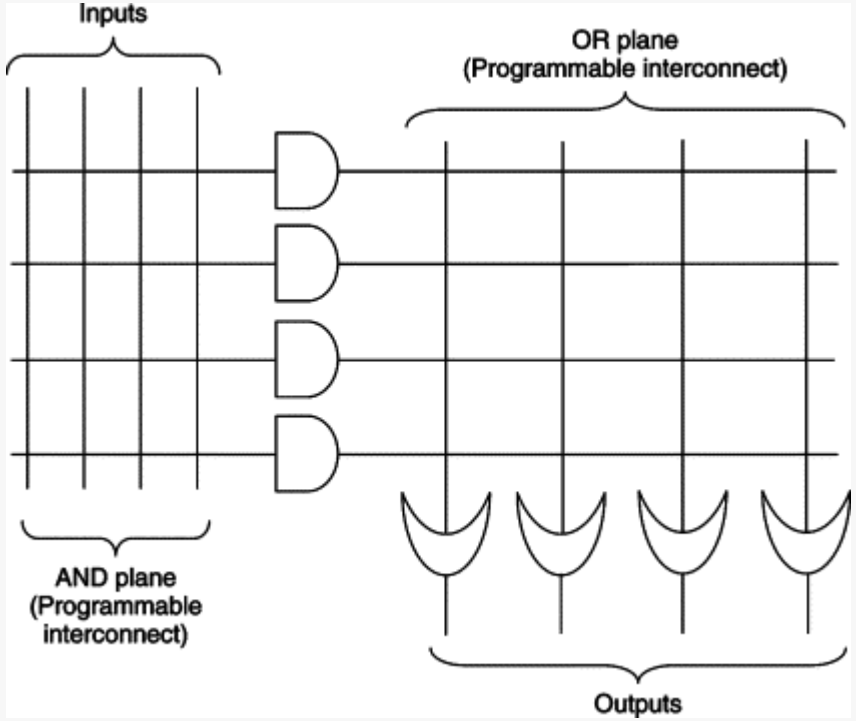


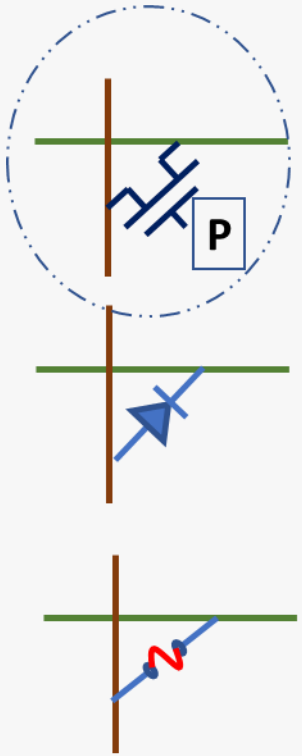


https://www.elprocus.com/what-are-pal-and-pla-design-and-differences/

For Exercise

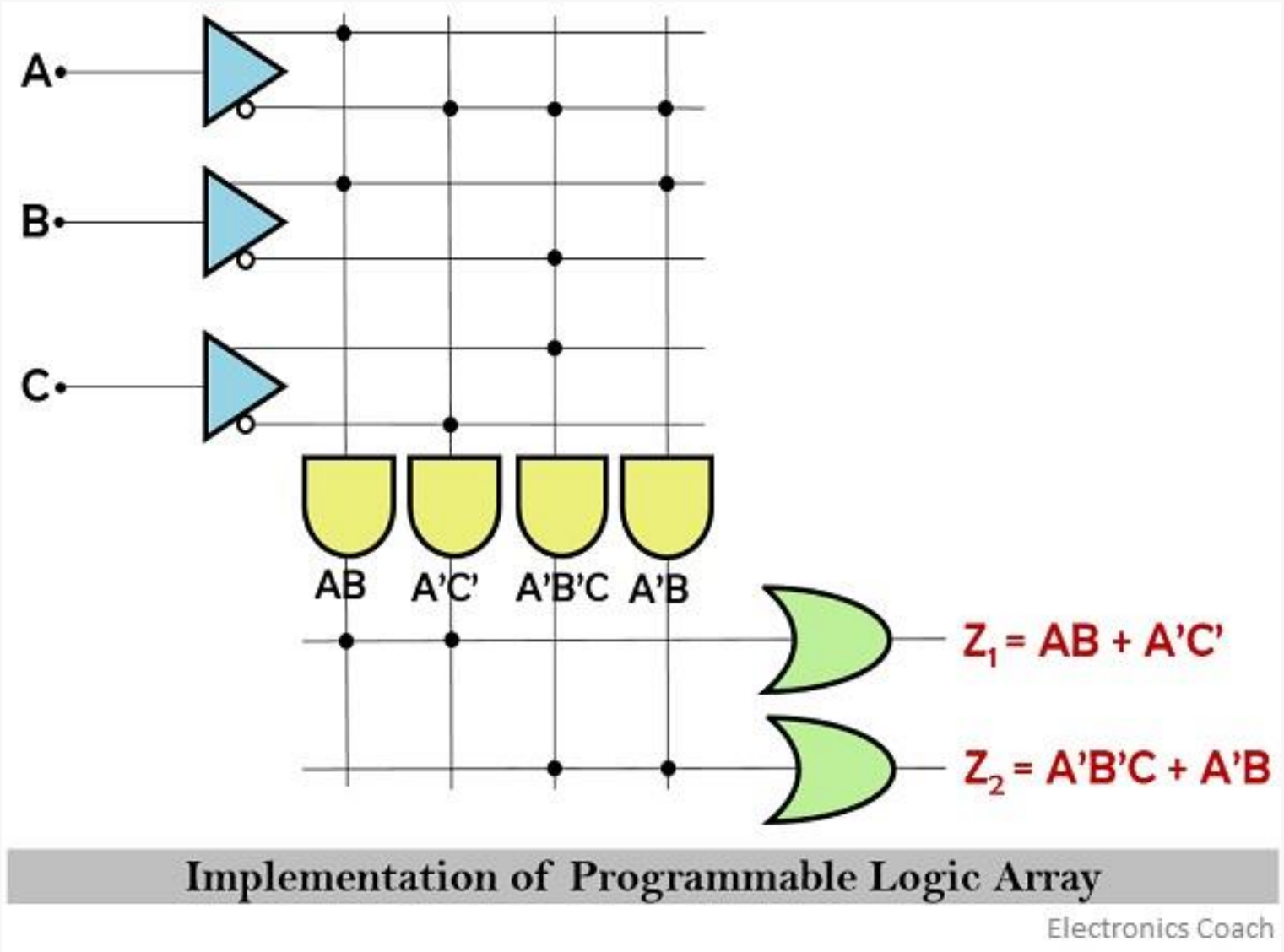






How to Configure?

Example of PLA

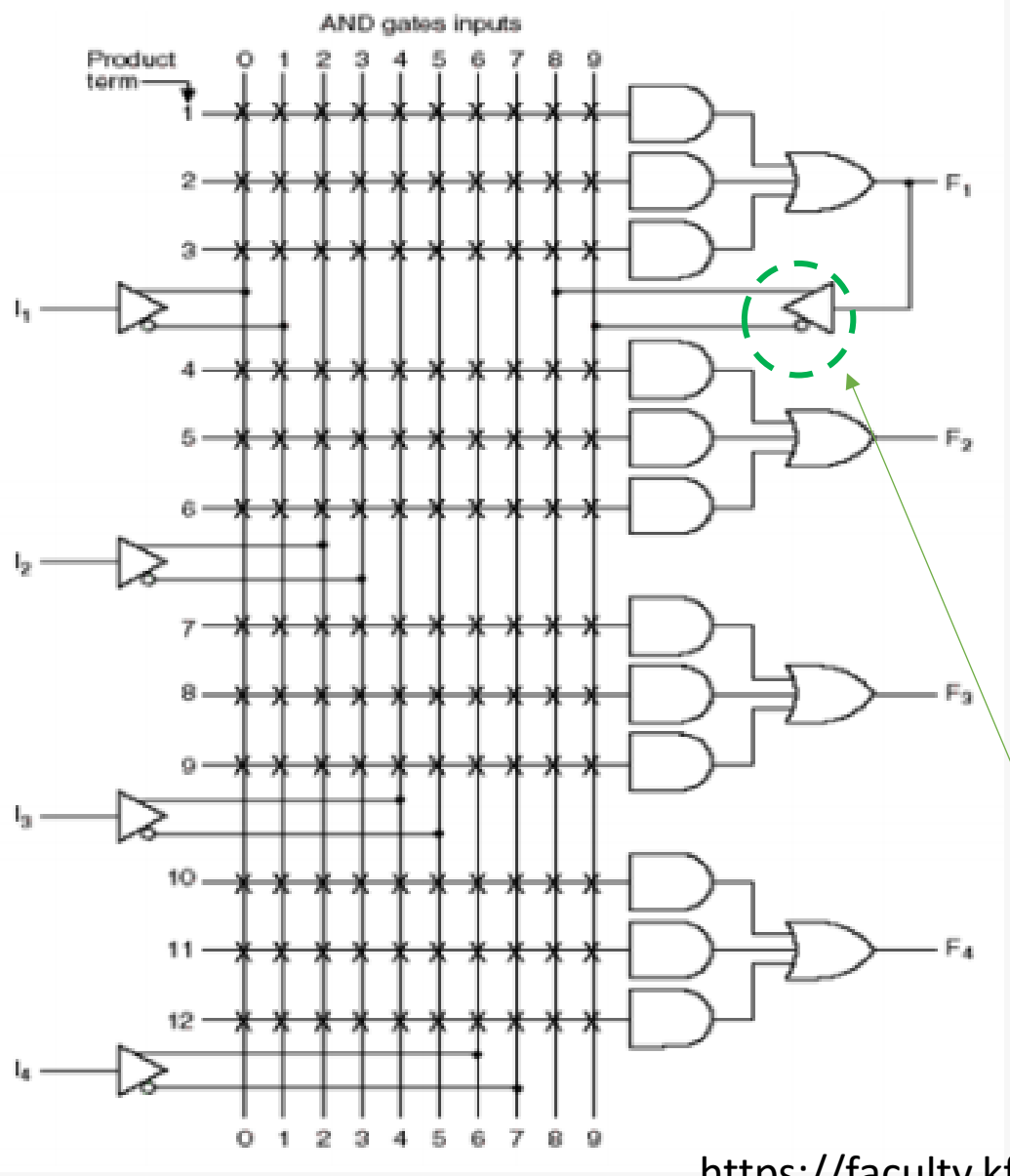


Implementation of Programmable Logic Array

Electronics Coach

https://electronicscoach.com/programmable-logic-array.html

PAL – Programmable Array Logic Example



- PAL Device is a PLD with fixed OR array and a programmable AND array
- As only AND gates are programmable, PAL is easier to program

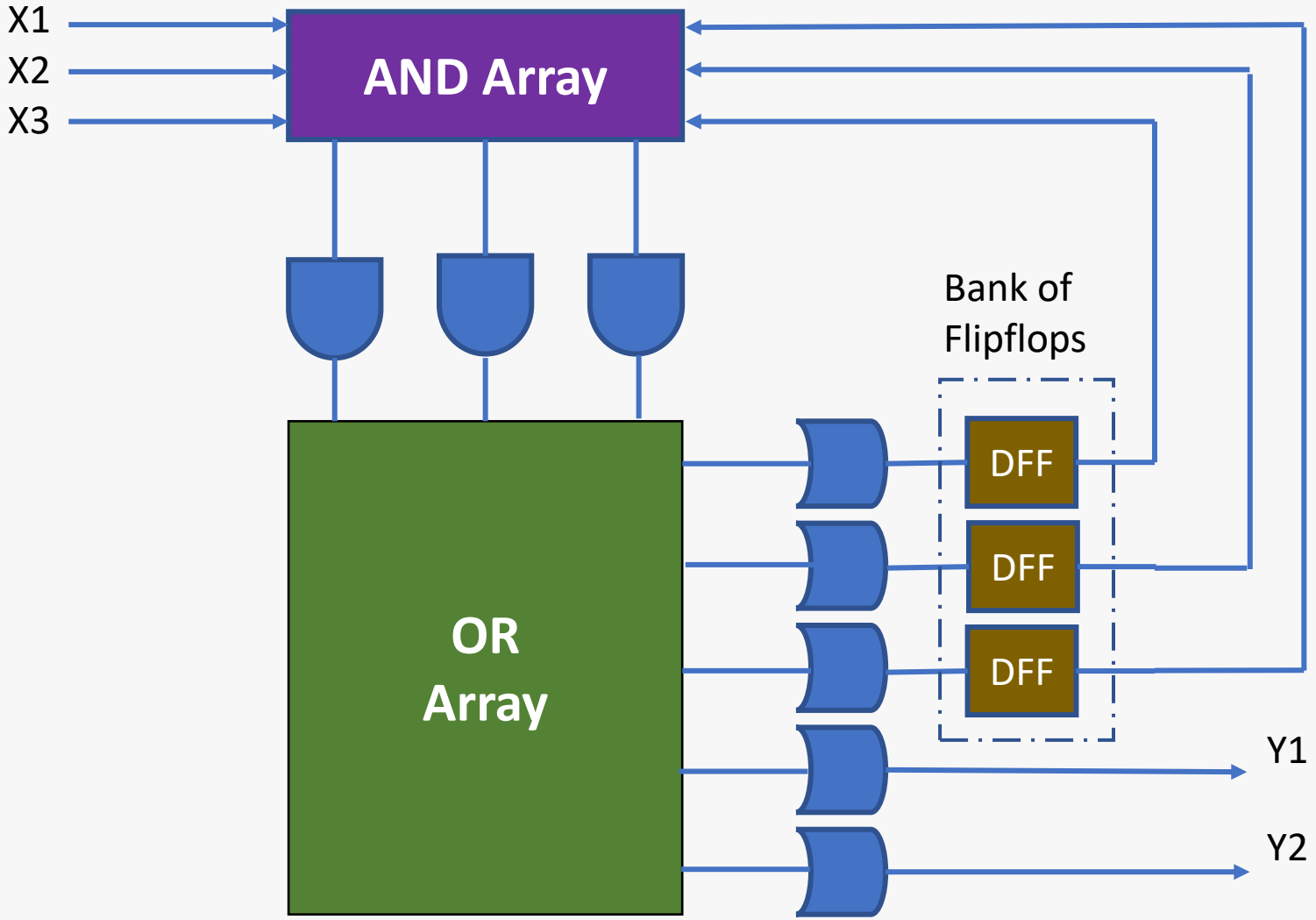
but it is less flexible compared to PLA devices

Example device:

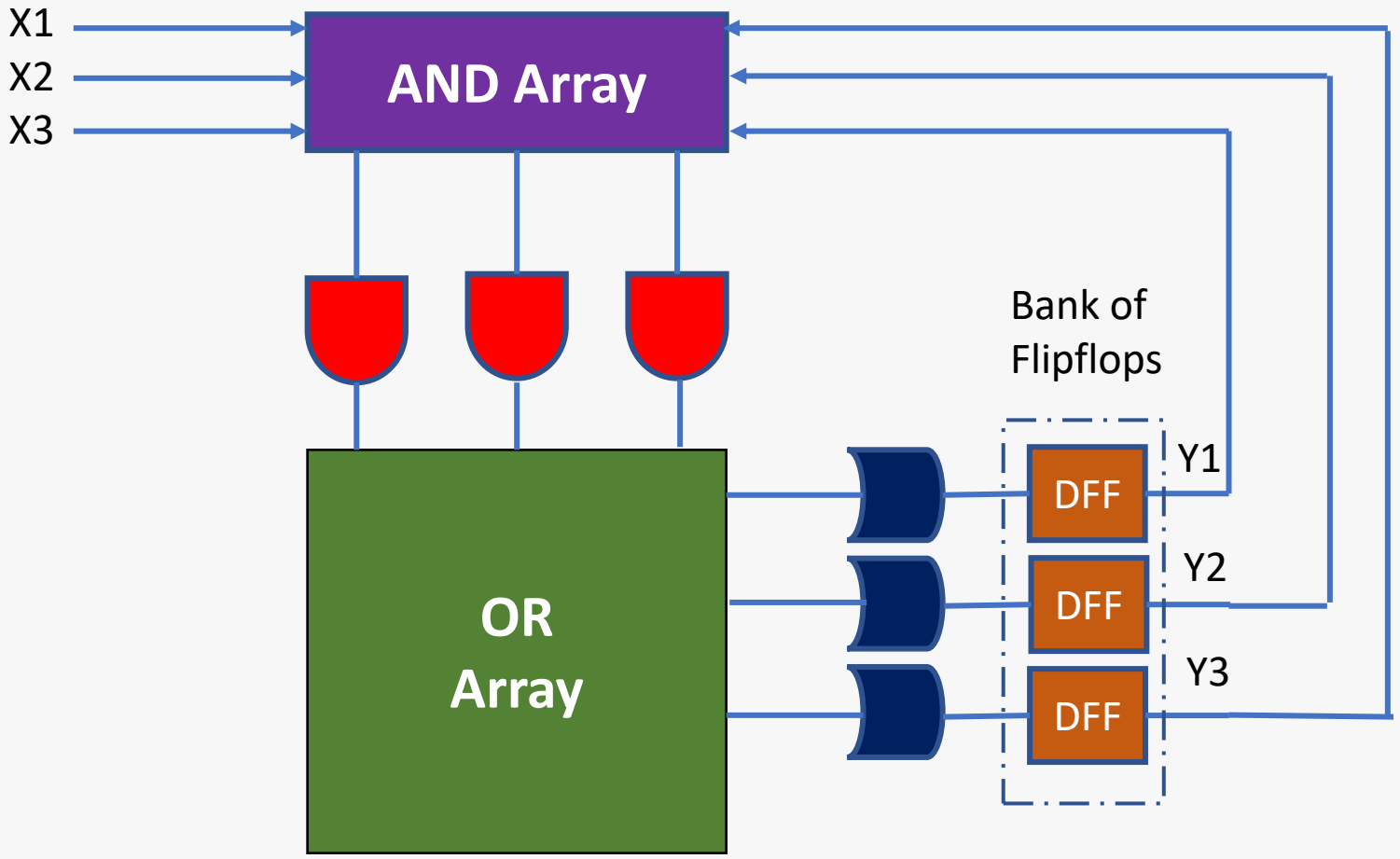
- This device has 4 inputs and 4 outputs.
- Each input has a buffer and inverter, output is from fixed OR gate
- The device has 4 sections: Each section comprises 3 wide AND-OR array, meaning three programmable AND arrays in each section
- Each AND gates has 10 programmable input connections
- One of the outputs (see F1) can be fed back to the inputs of AND gate through programmable connections

https://faculty.kfupm.edu.sa/COE/aimane/coe202/PLDs.pdf

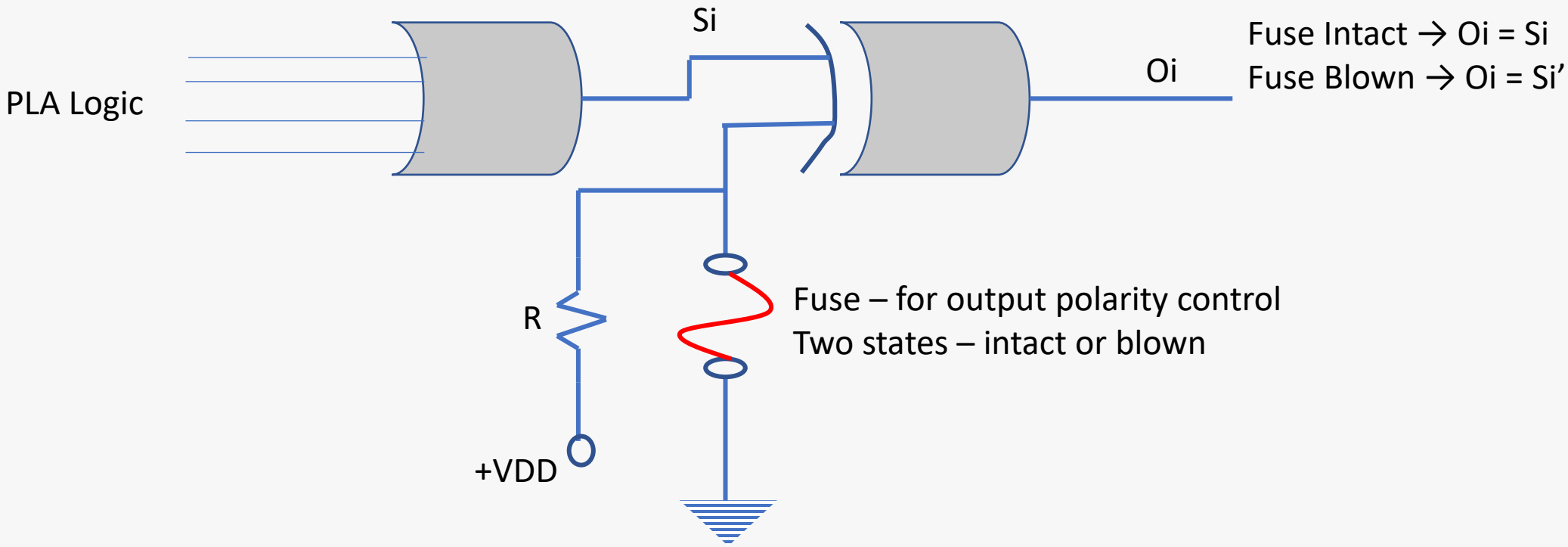
Sequential Circuits with PLD – Mealy Machine



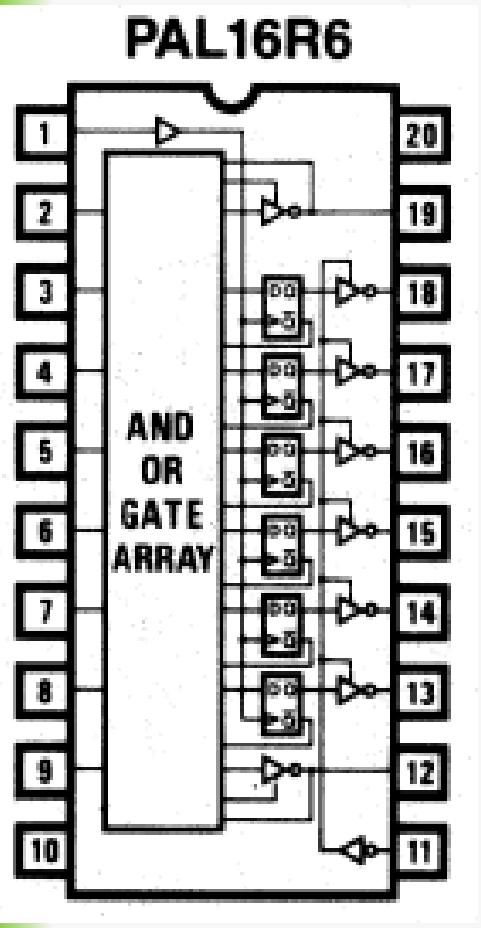
Sequential Circuits with PLD – Moore Machine

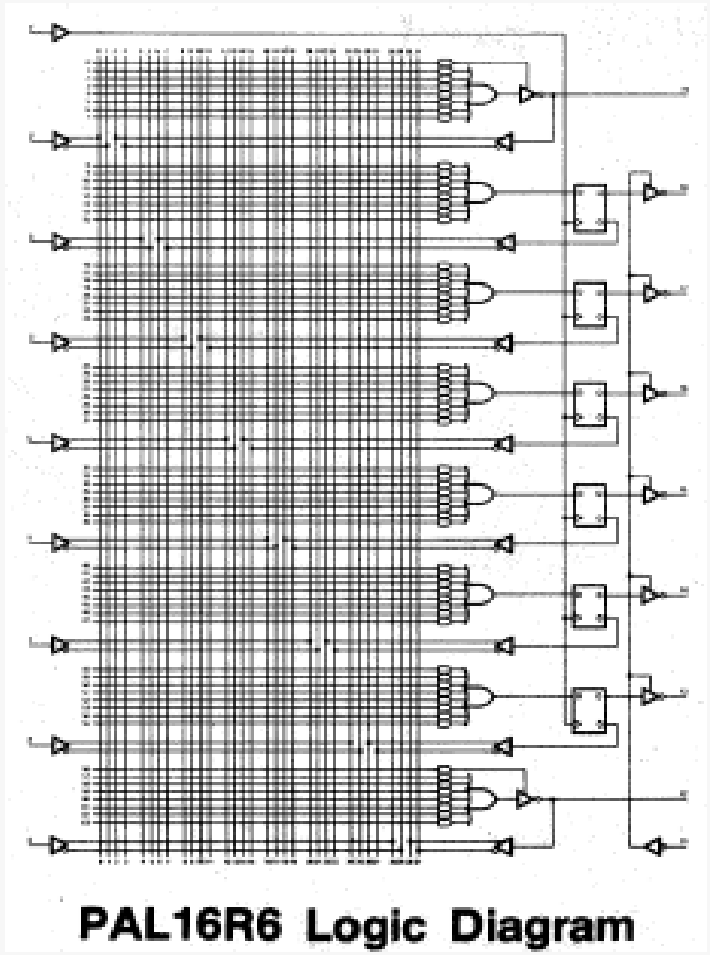


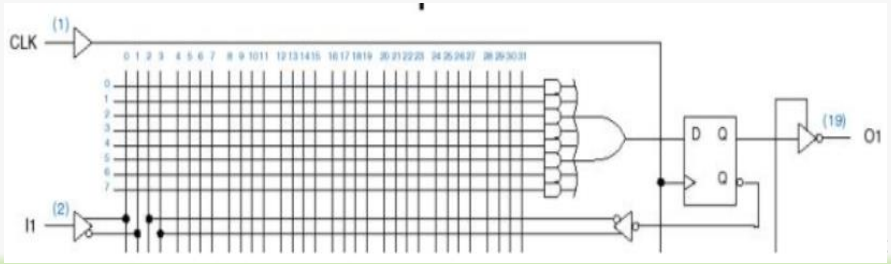
Output Polarity Control

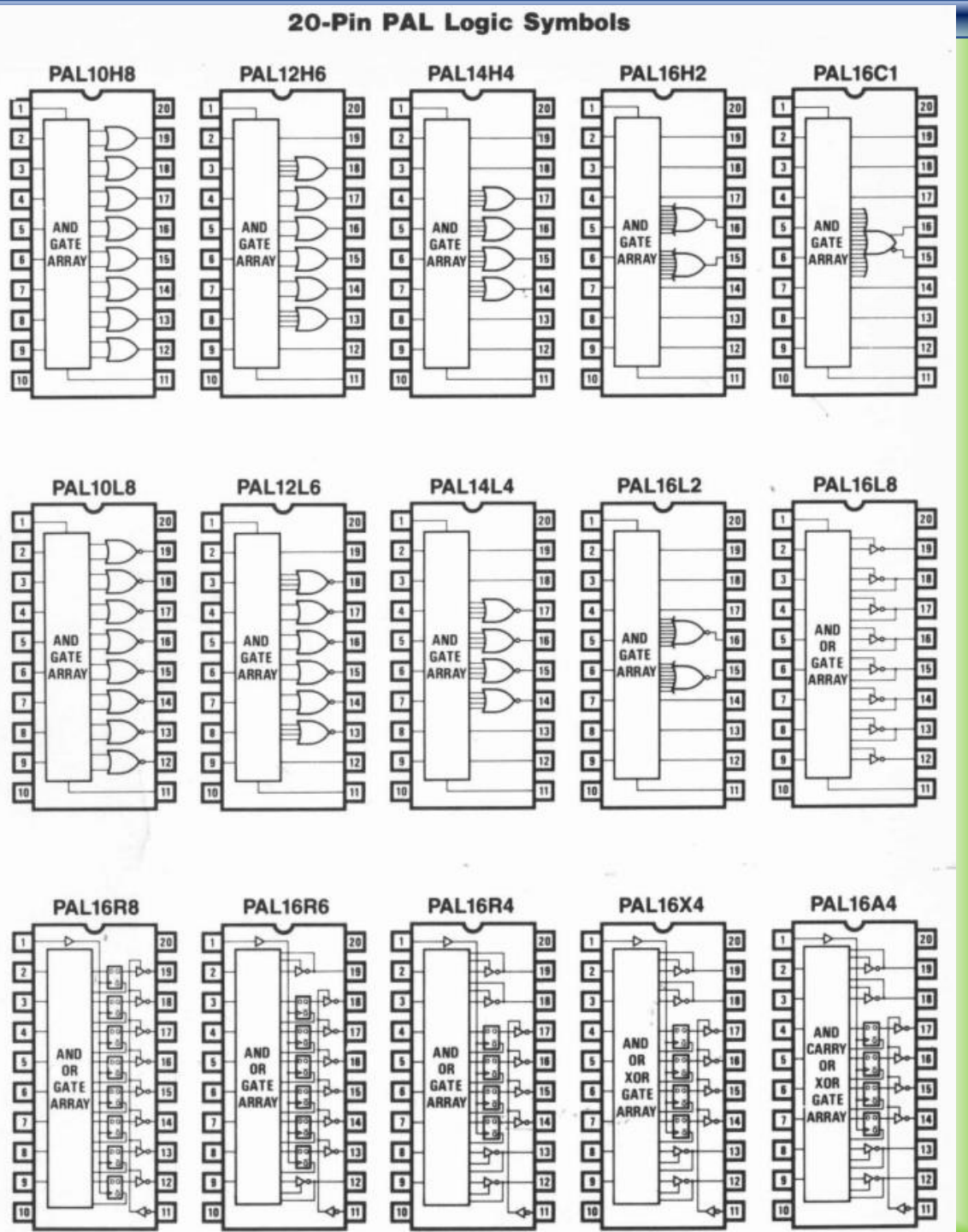


PLA Family Examples




PAL16R6 Logic Diagram





Lecture 19

EE 421 / CS 425

Digital System Design

Fall 2026

Shahid Masud

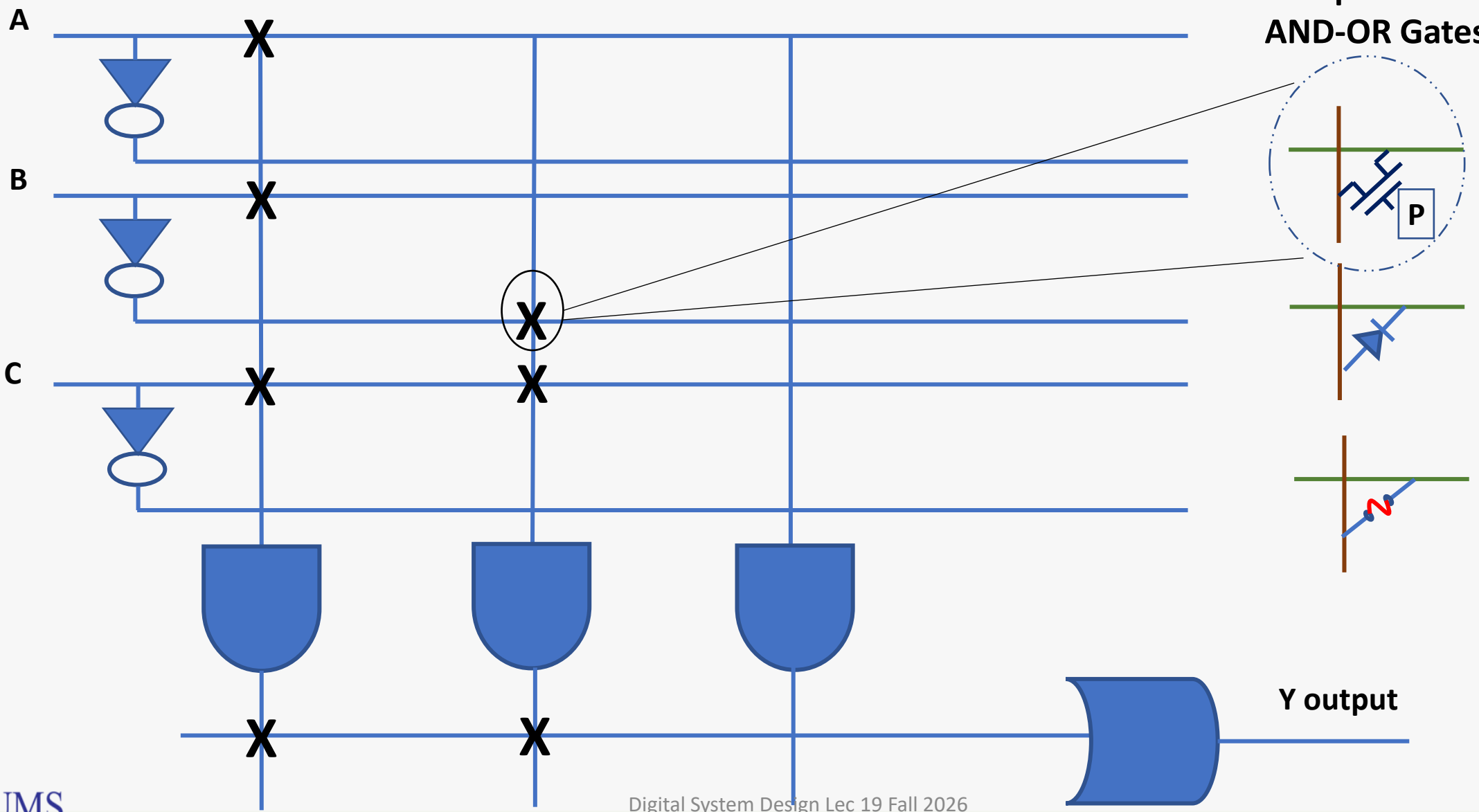
Topics

- PAL, PLA, Examples, Polarity Control Circuit in I/O (Recap)
- Datasheet of PLA/PAL
- Example of an I/O Block for SPLD
- Field Programming technology
- CPLD Architecture
- CPLD Examples from Xilinx and Altera
- Logic Expander – Shareable and Parallel
- Output Block of PLD
- FPGA Architecture

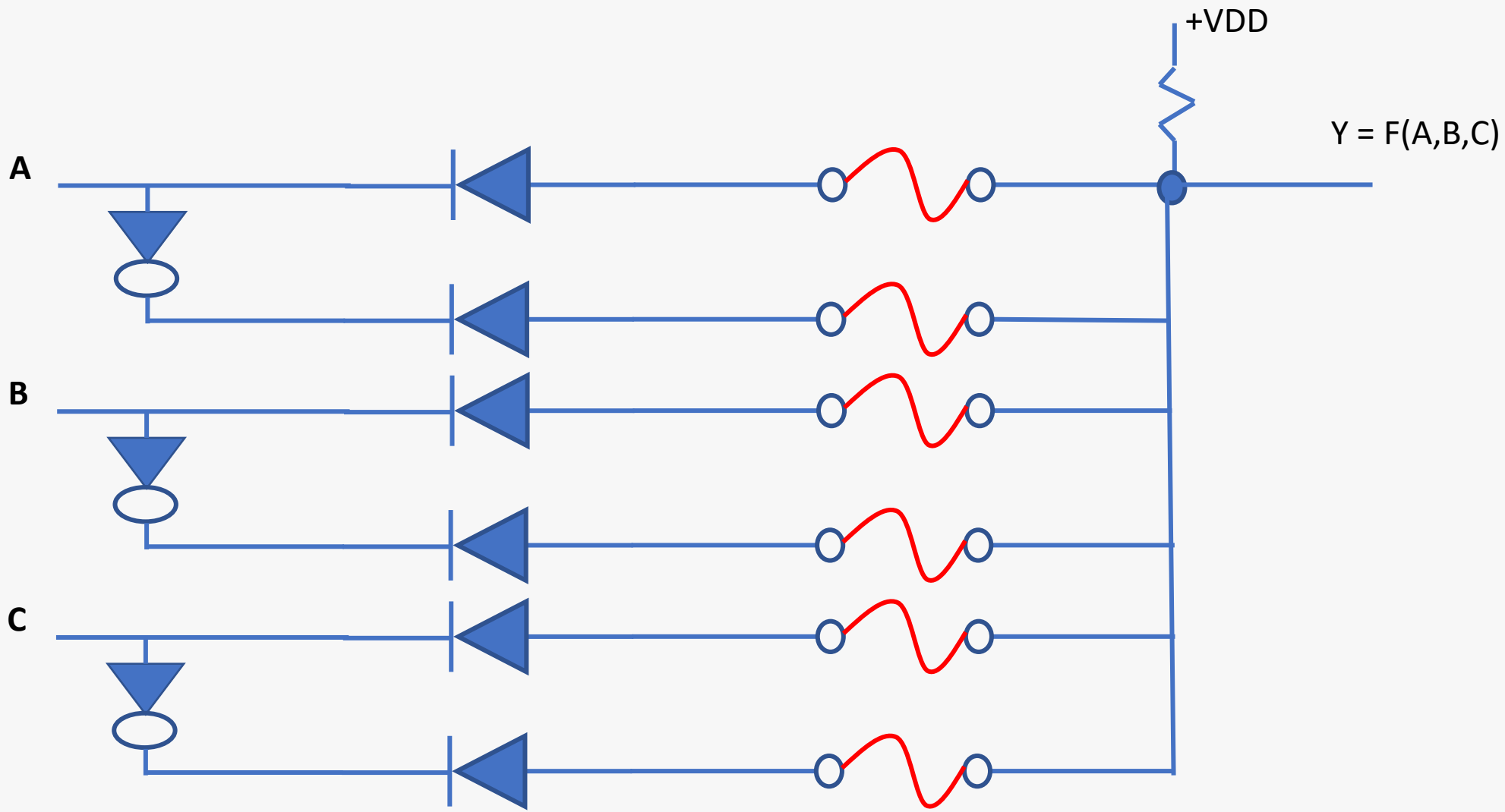
QUIZ NEXT Tuesday

AND-OR Logic Array

PLA = Programmable Logic Array Comprises Programmable AND-OR Gates

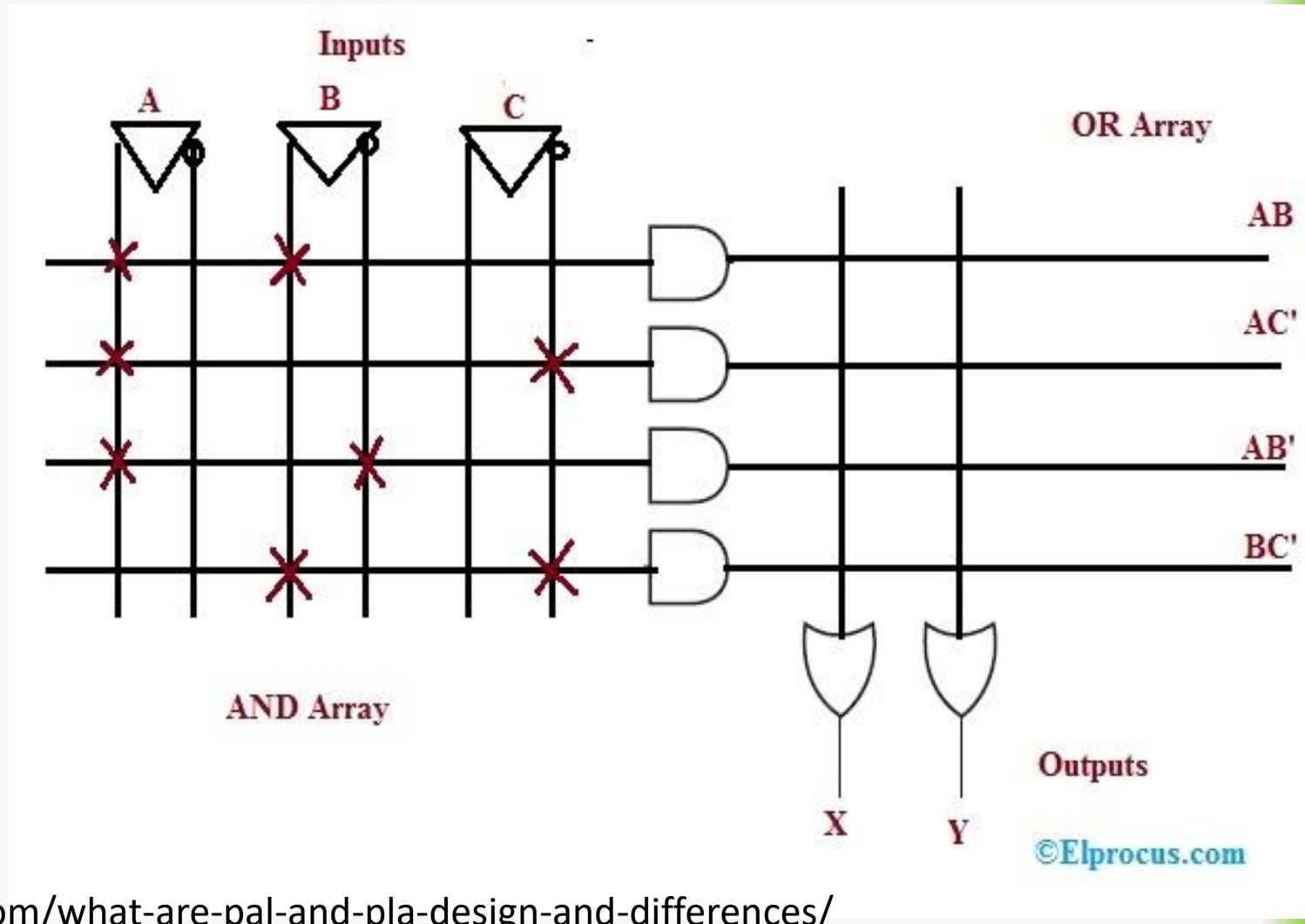


Fuse Programmable Logic Array



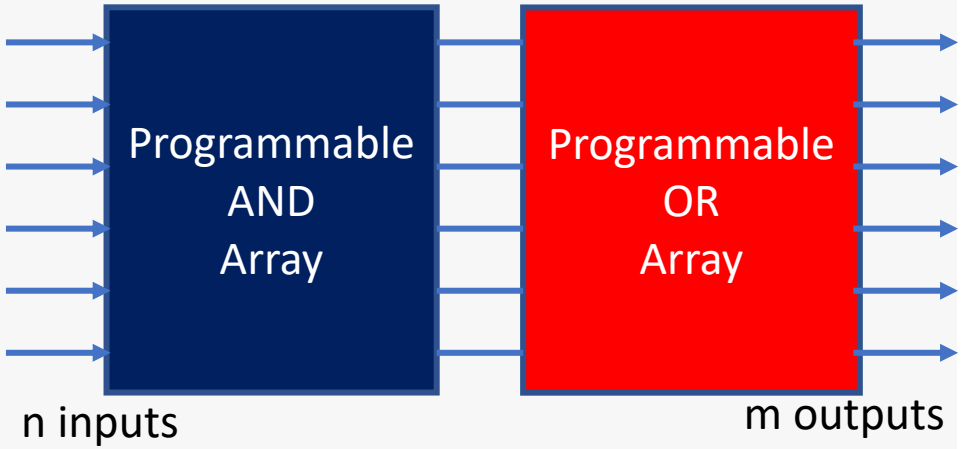
PAL – Programmable Array Logic

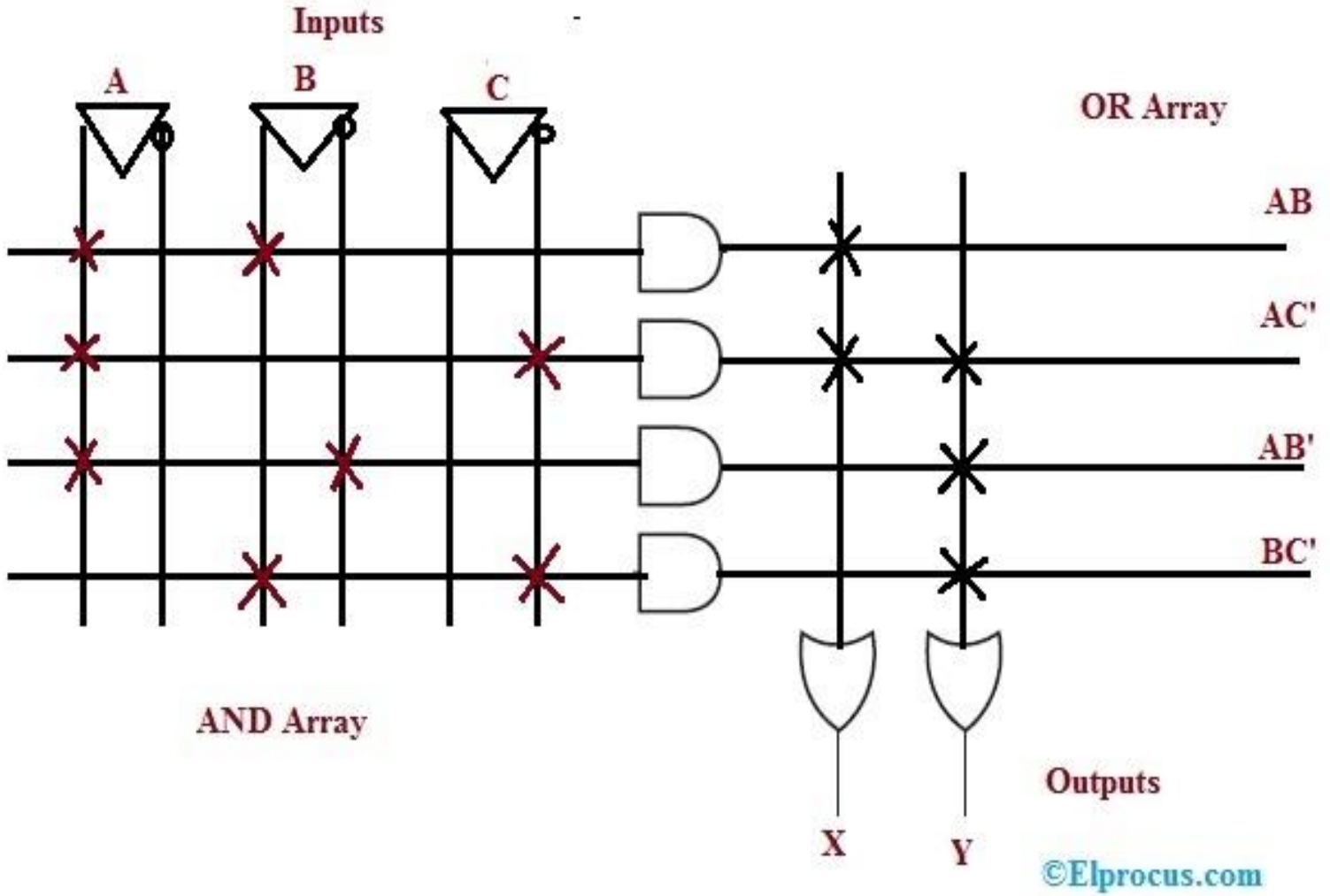




https://www.elprocus.com/what-are-pal-and-pla-design-and-differences/

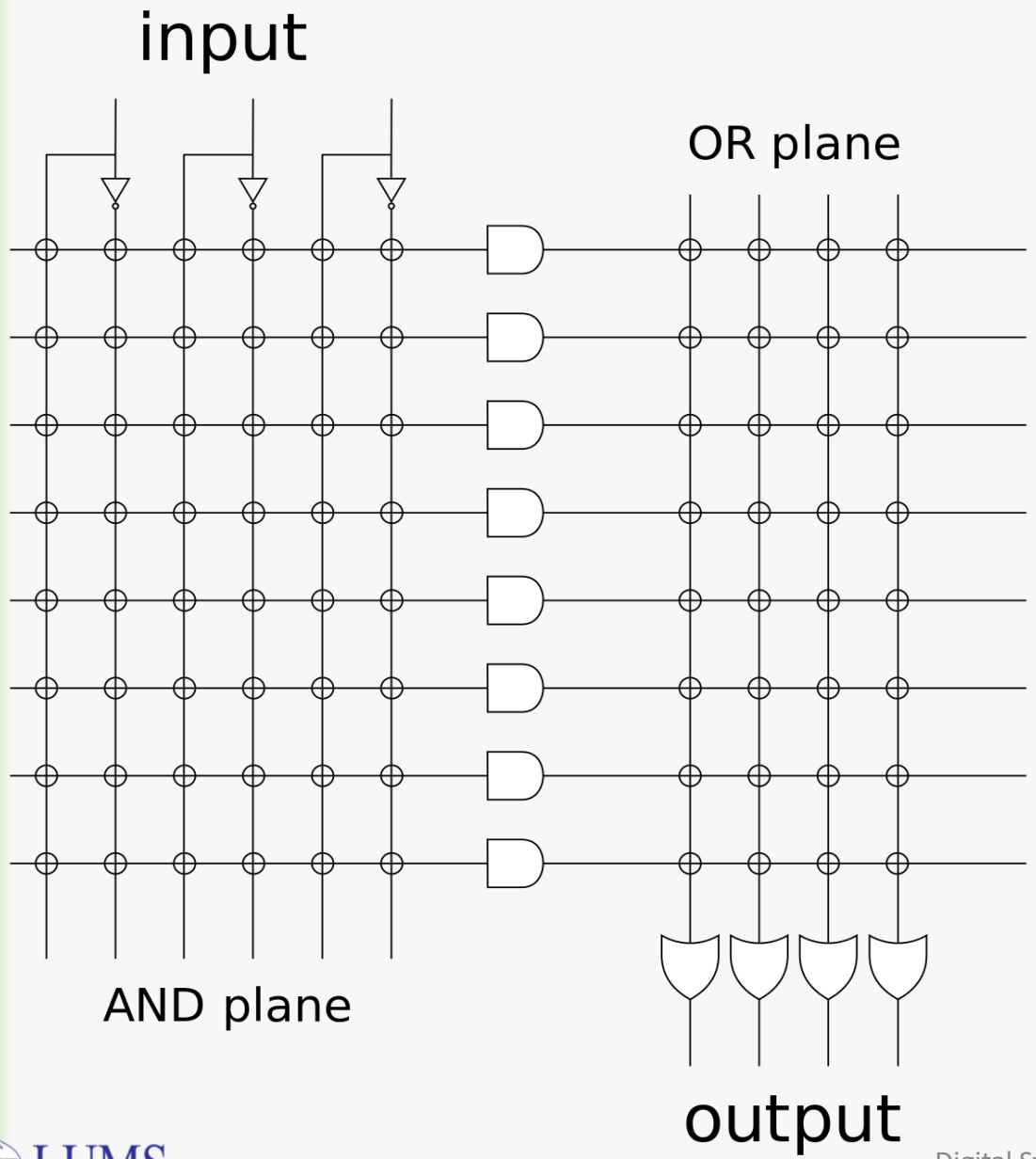
PLA – Programmable Logic Array

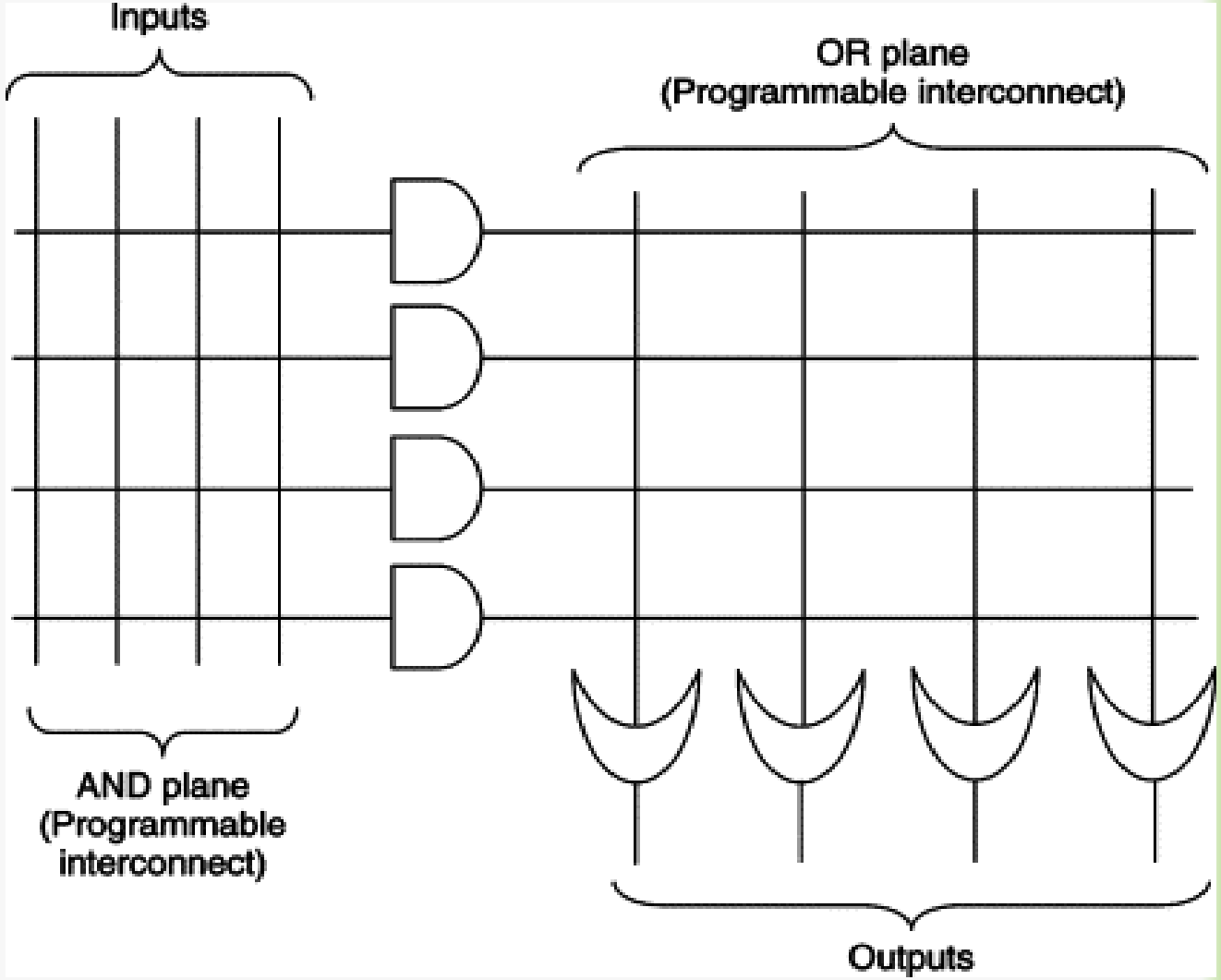




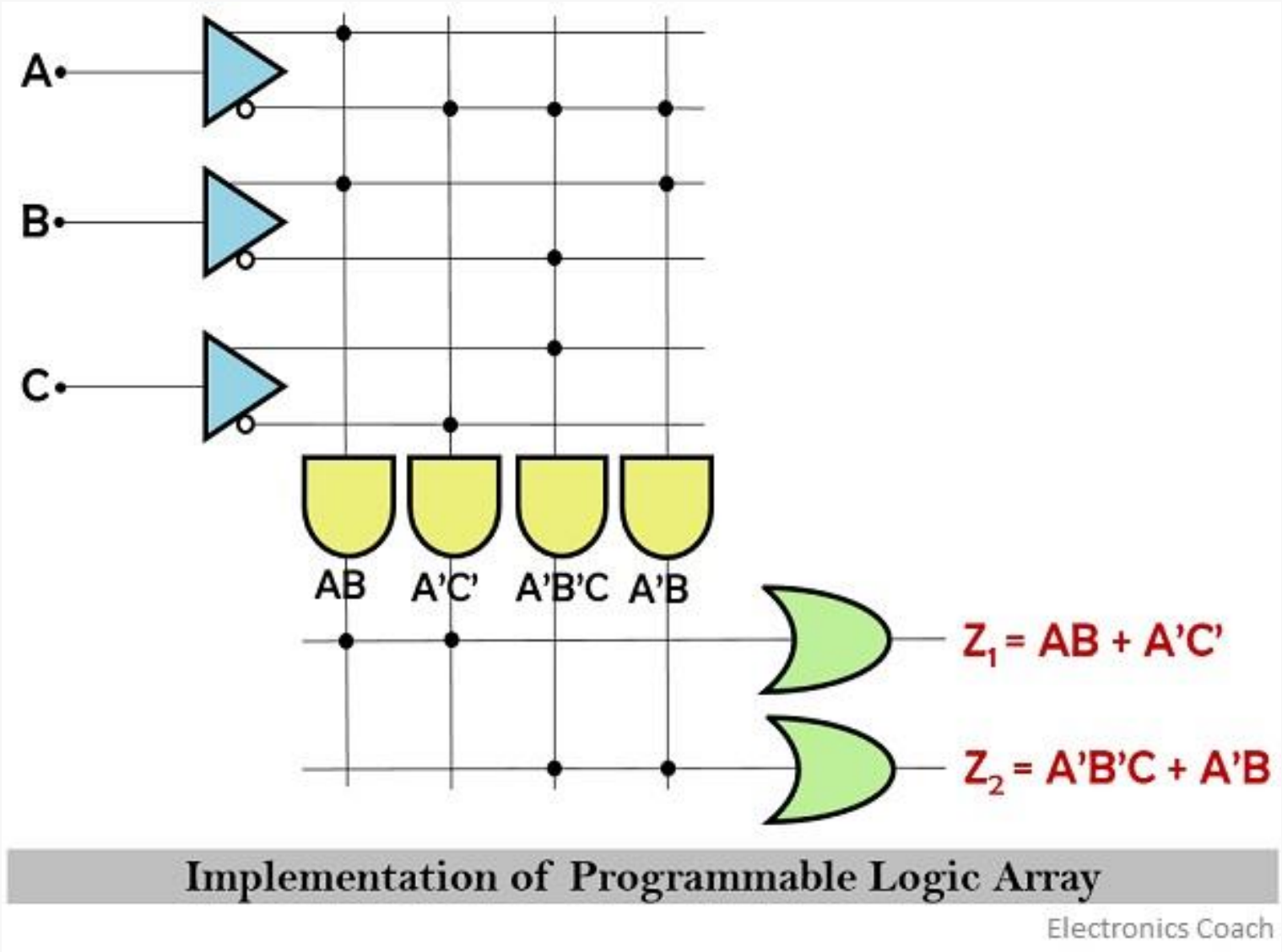
https://www.elprocus.com/what-are-pal-and-pla-design-and-differences/

For Exercise



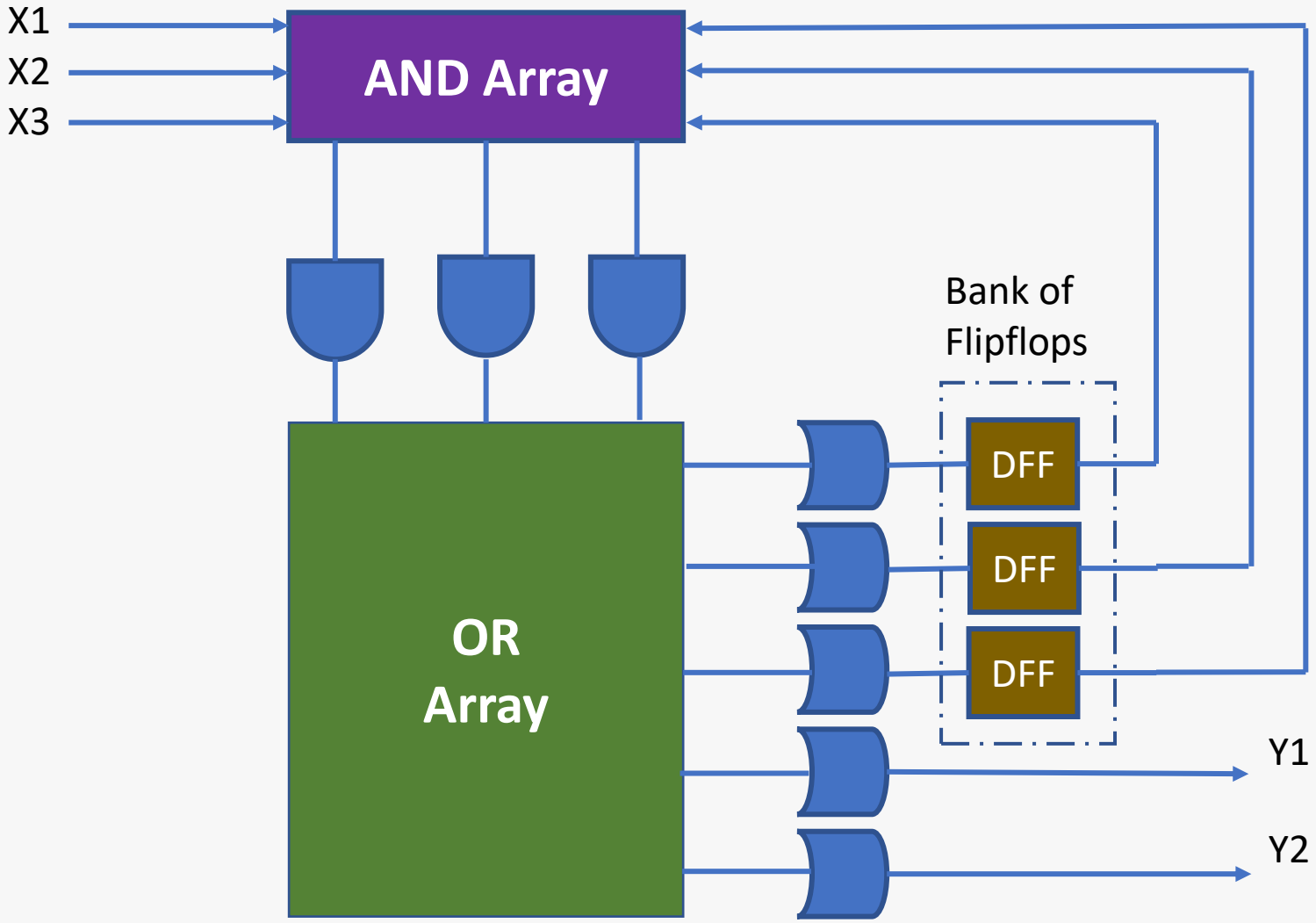


Example of PLA

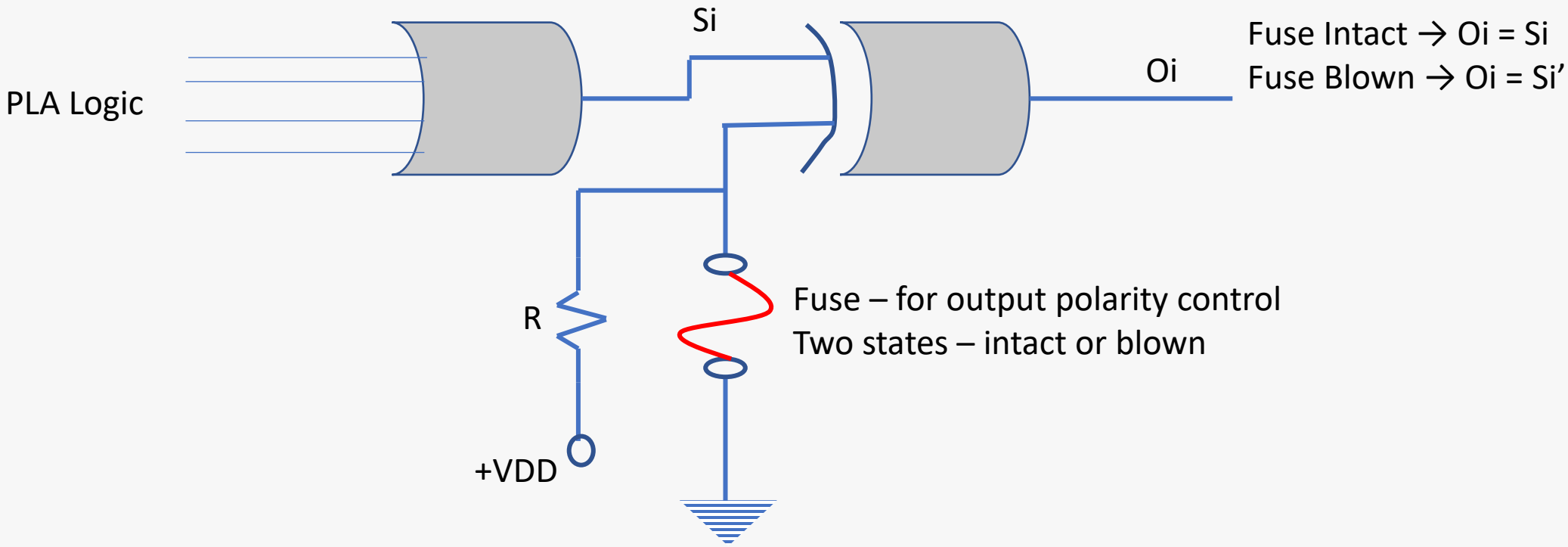


https://electronicscoach.com/programmable-logic-array.html

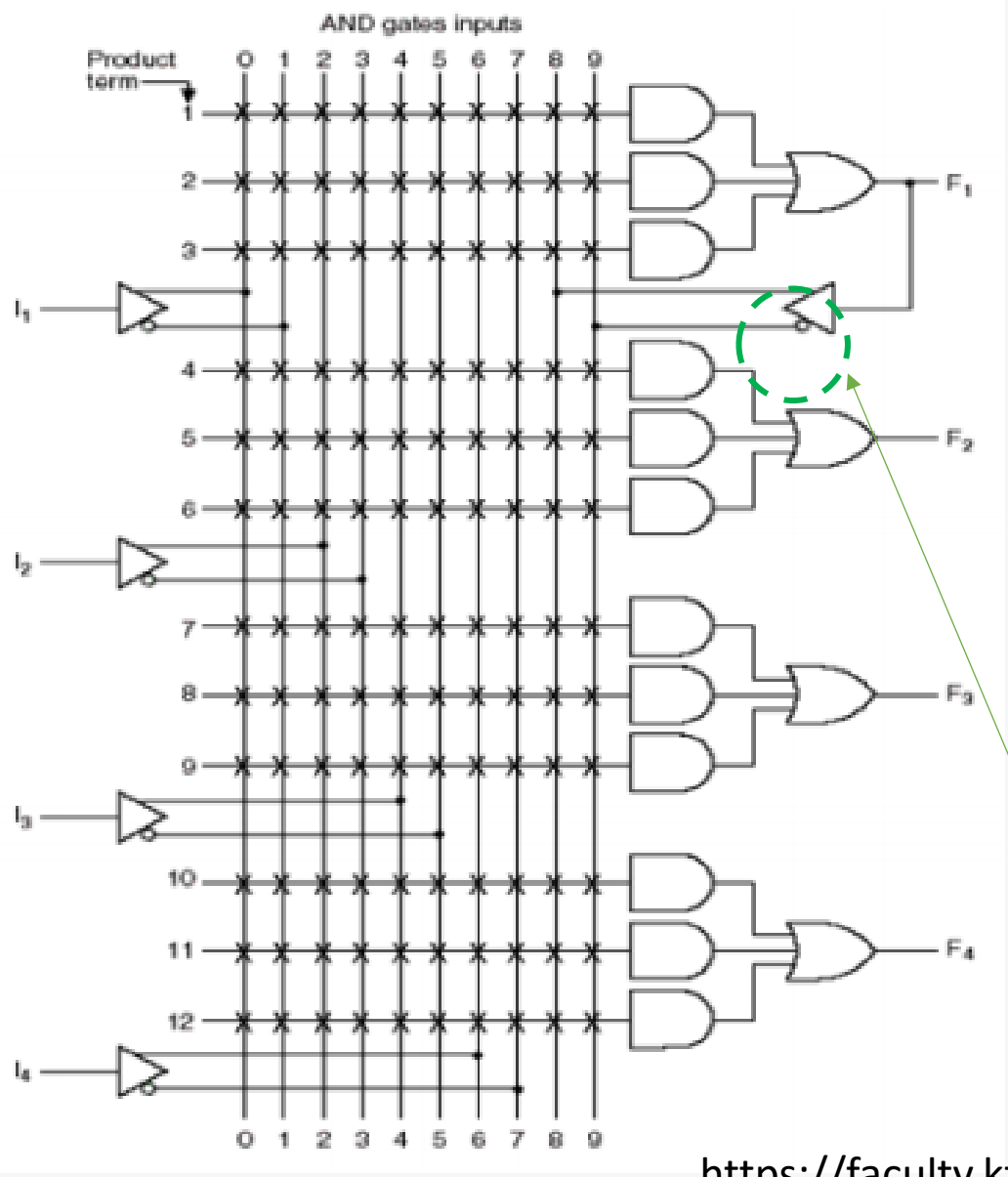
Sequential Circuits with PLD – Mealy Machine



Output Polarity Control



PAL – Programmable Array Logic Example



- PAL Device is a PLD with fixed OR array and a programmable AND array
- As only AND gates are programmable, PAL is easier to program

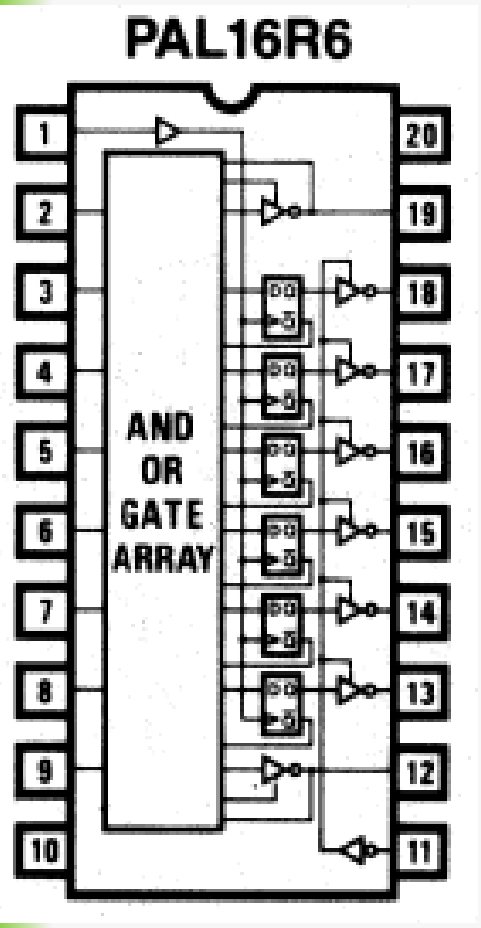
but it is less flexible compared to PLA devices

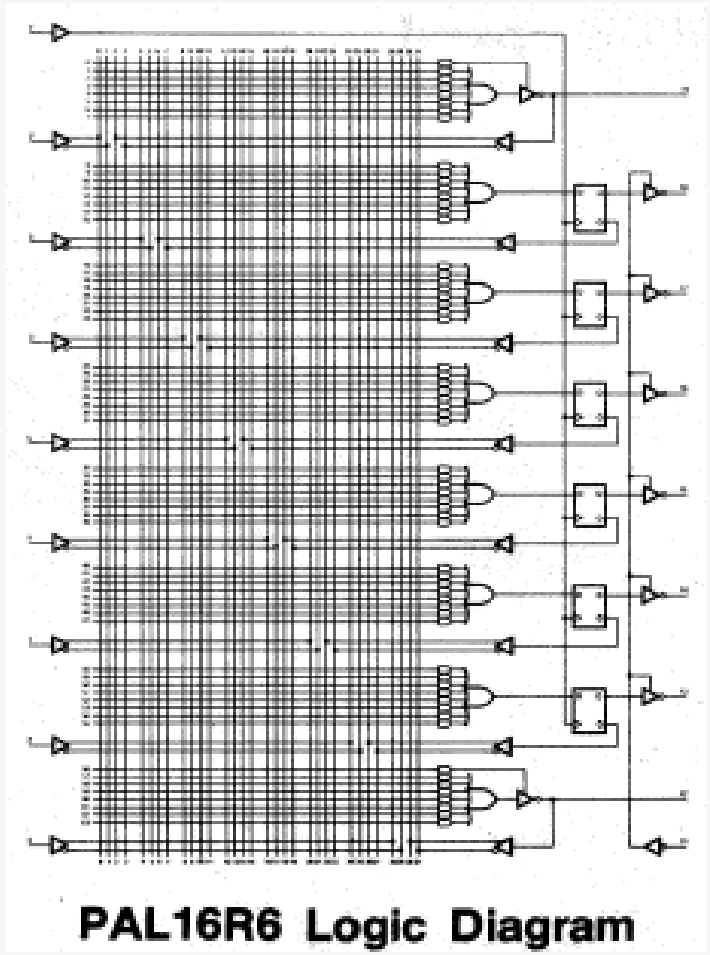
Example device:

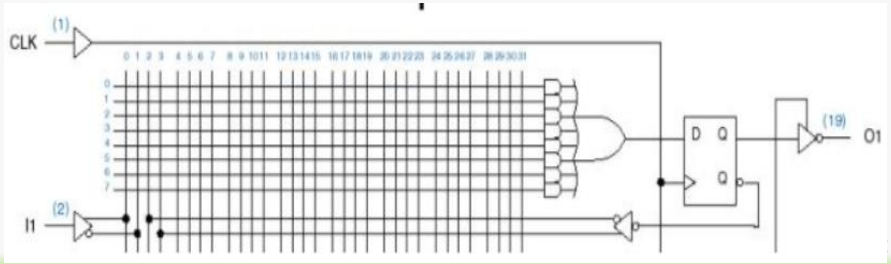
- This device has 4 inputs and 4 outputs.
- Each input has a buffer and inverter, output is from fixed OR gate
- The device has 4 sections: Each section comprises 3 wide AND-OR array, meaning three programmable AND arrays in each section
- Each AND gates has 10 programmable input connections
- One of the outputs (see F1) can be fed back to the inputs of AND gate through programmable connections

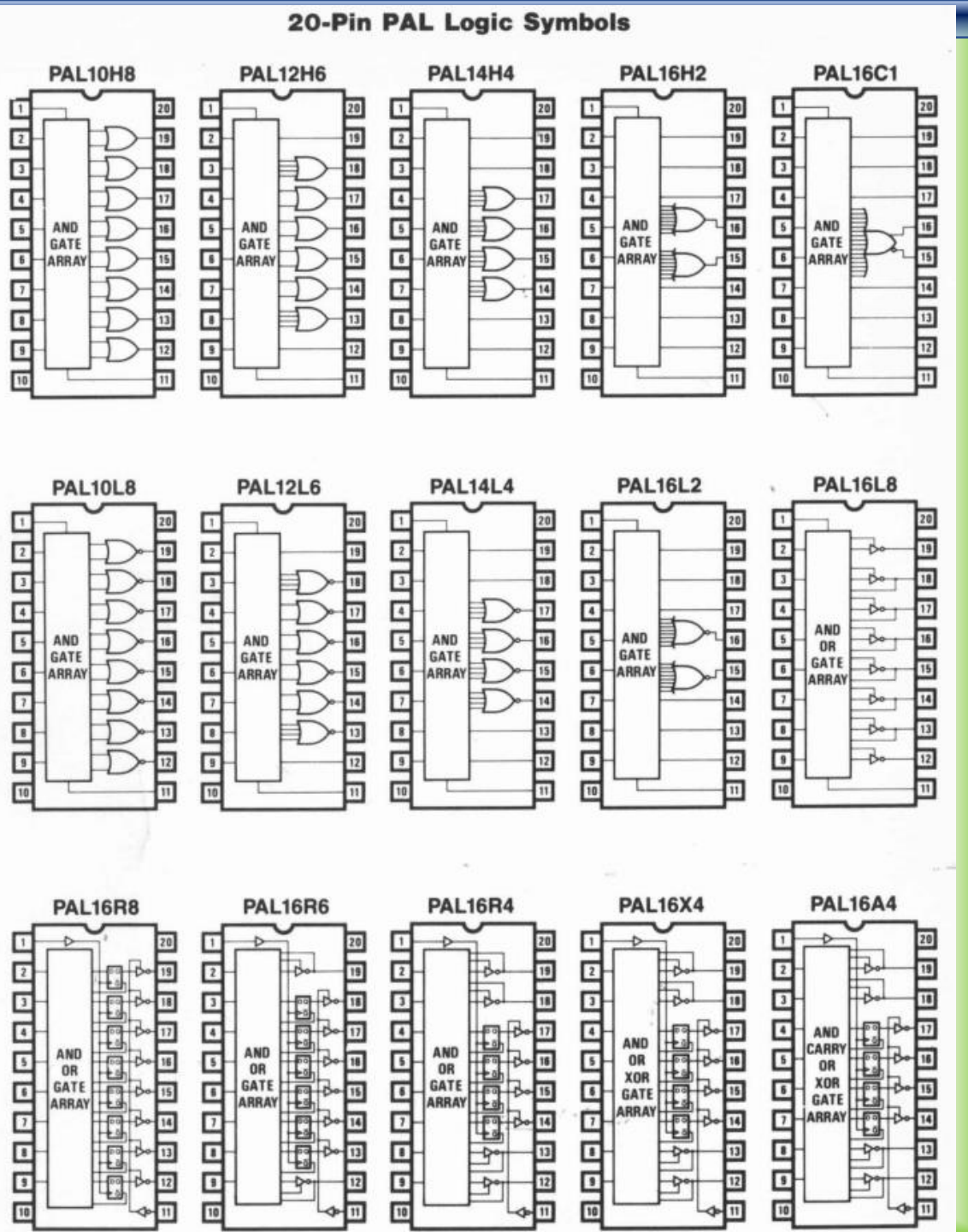
https://faculty.kfupm.edu.sa/COE/aimane/coe202/PLDs.pdf

PLA Family Examples




PAL16R6 Logic Diagram

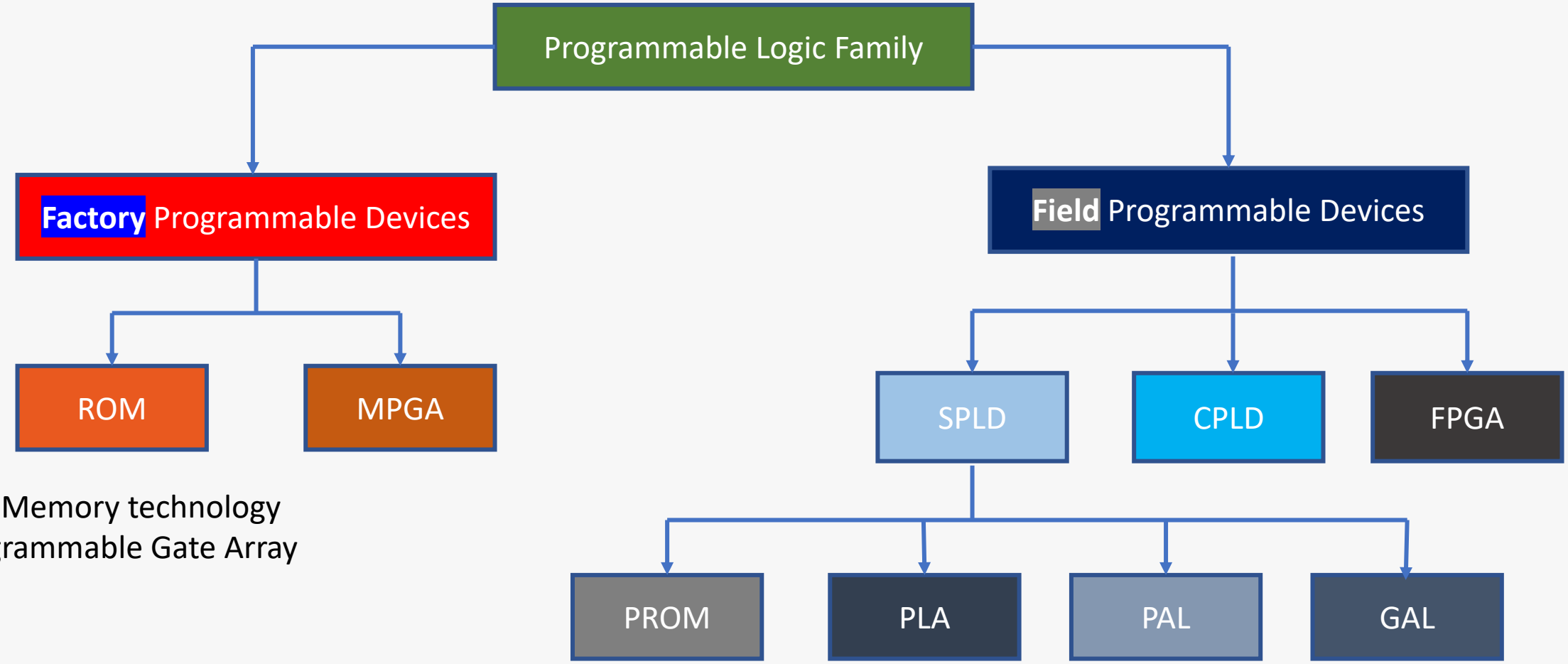




Datasheet of PLA / PAL

- Go through PAL and PLA Datasheet for some examples

Programmable Logic Broad Classification



Read only Memory technology
Mask Programmable Gate Array

Simple and Complex Programmable Logic Devices
Field Programmable Gate Array
Generic Array Logic (complex PAL by lattice semi), Programmable Logic Array

Programmable Hardware Devices Comparison

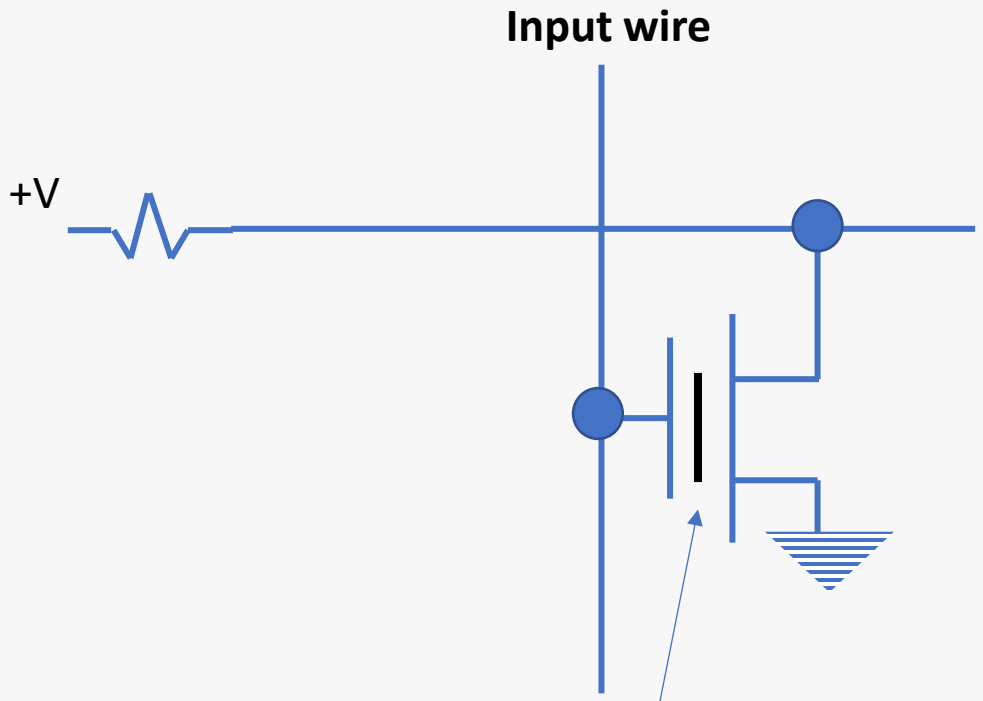
Table 3-1: *A Comparison of Programmable Devices*

	SPLD	CPLD	FPGA
Density	Low Few hundred gates	Low to Medium 500 to 12,000 gates	Medium to High 3,000 to 5,000,000 gates
Timing	Predictable	Predictable	Unpredictable
Cost	Low	Low to Medium	Medium to High
Major Vendors	Lattice Semiconductor Cypress AMD	Xilinx Altera	Xilinx Altera Lattice Semiconductor Actel
Example Device Families	**Lattice Semiconductor** GAL16LV8 GAL22V10 **Cypress** PALCE16V8 **AMD** 22V10	**Xilinx** CoolRunner XC9500 **Altera** MAX	**Xilinx** Virtex Spartan **Altera** Stratix **Lattice** Mach ECP **Actel** Accelerator

Ref: Roth & John Book

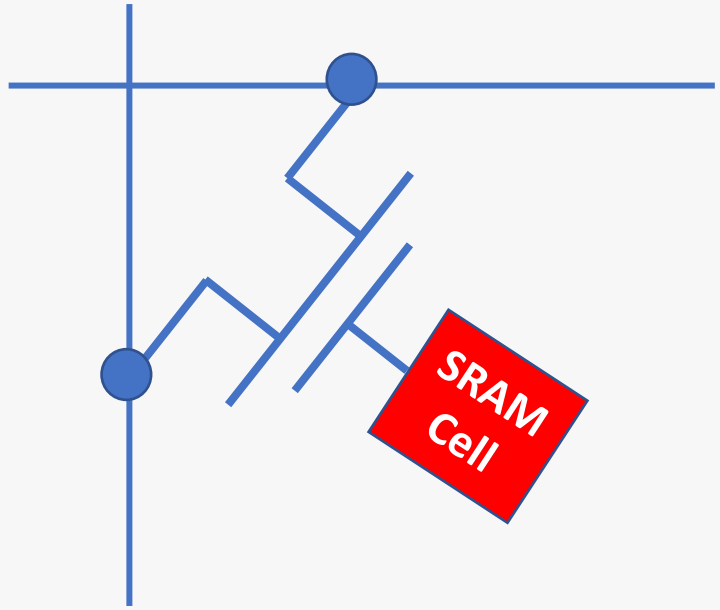
Interconnect Technology for Field Programming

EEPROM / FLASH



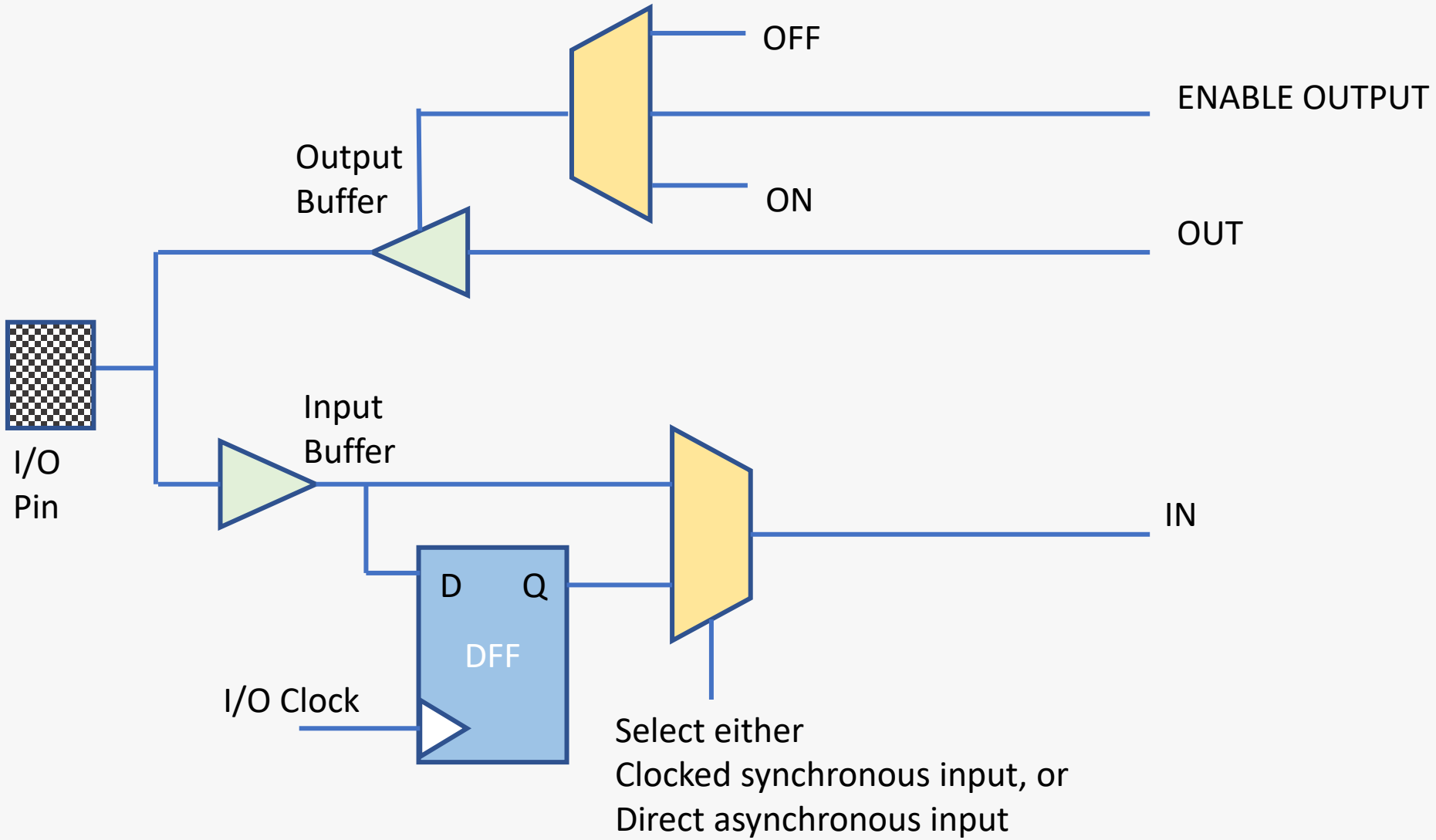
Floating Gate
Tunneling process
Used in EEPROM, USB Flash drive, etc.

SRAM



SRAM Cell is connected to Gate terminal
MOS Switch behaves as per stored value in SRAM

Concept of a Programmable I/O Block



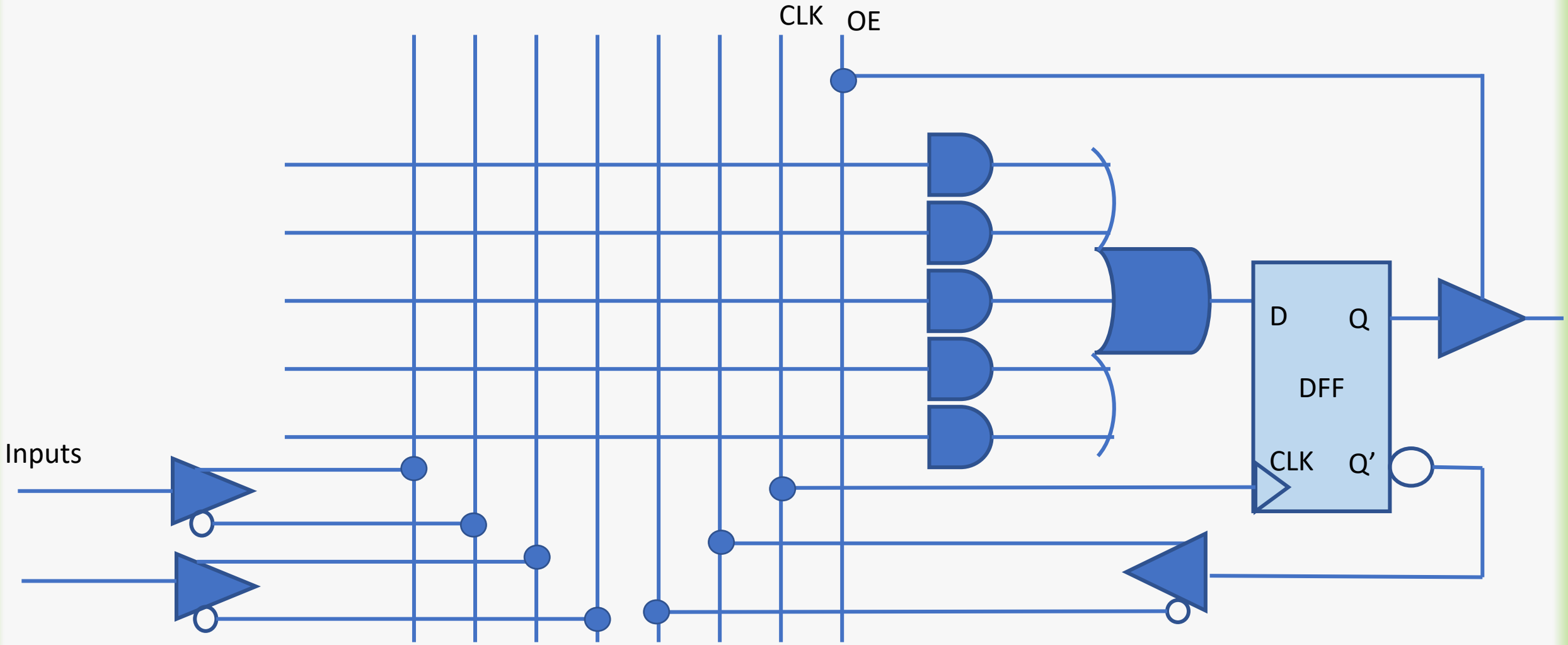
Introducing CPLD – Complex Programmable Logic Devices

Description of Basic (Registered) Macrocell

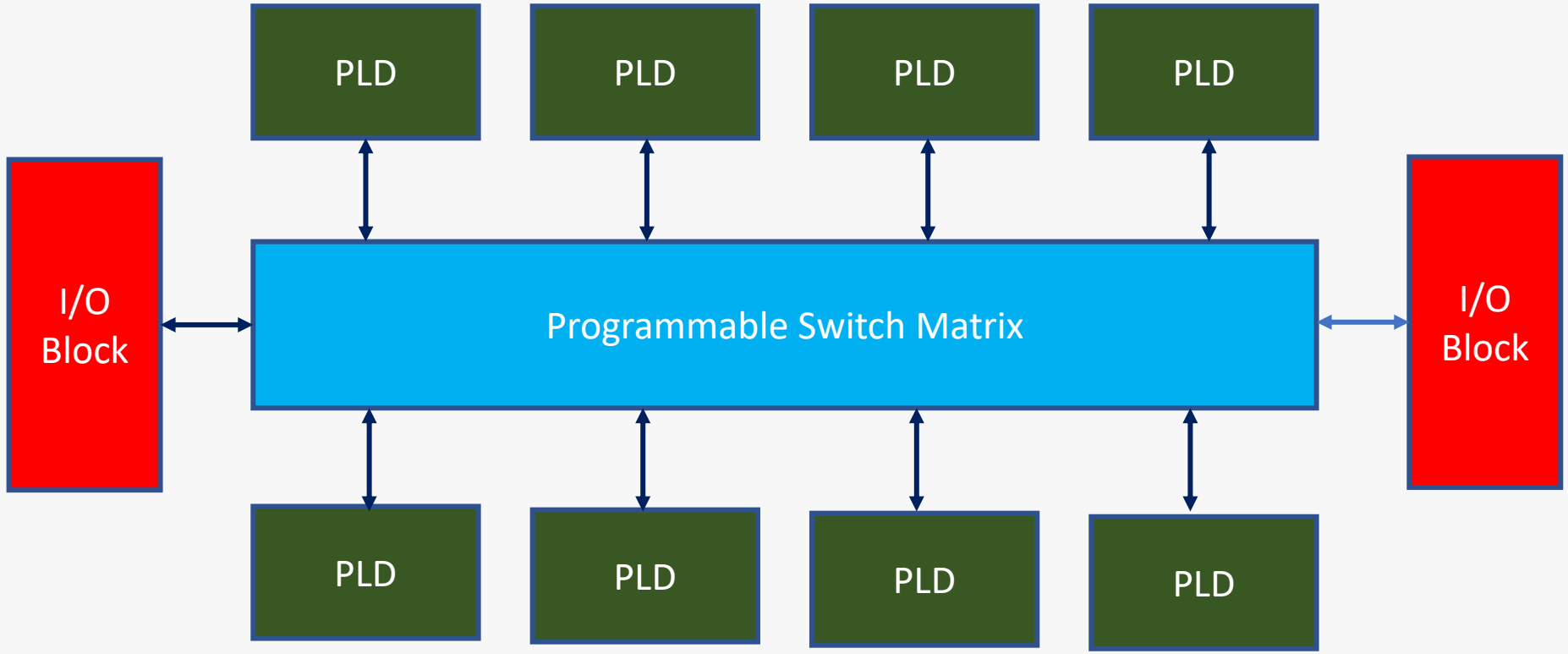
- Each section of an SPLD is called a macrocell that contains SOP combinational logic and an optional Flipflop
- A typical SPLD has from 8 to 10 macrocells within one chip
- All Flipflops are connected to common CLK input
- All three-state buffers at Flipflop output are controlled by a common OE input

Basic Macrocell Logic

PAL that includes a DFF is called a Registered PAL



Generic CPLD Configuration



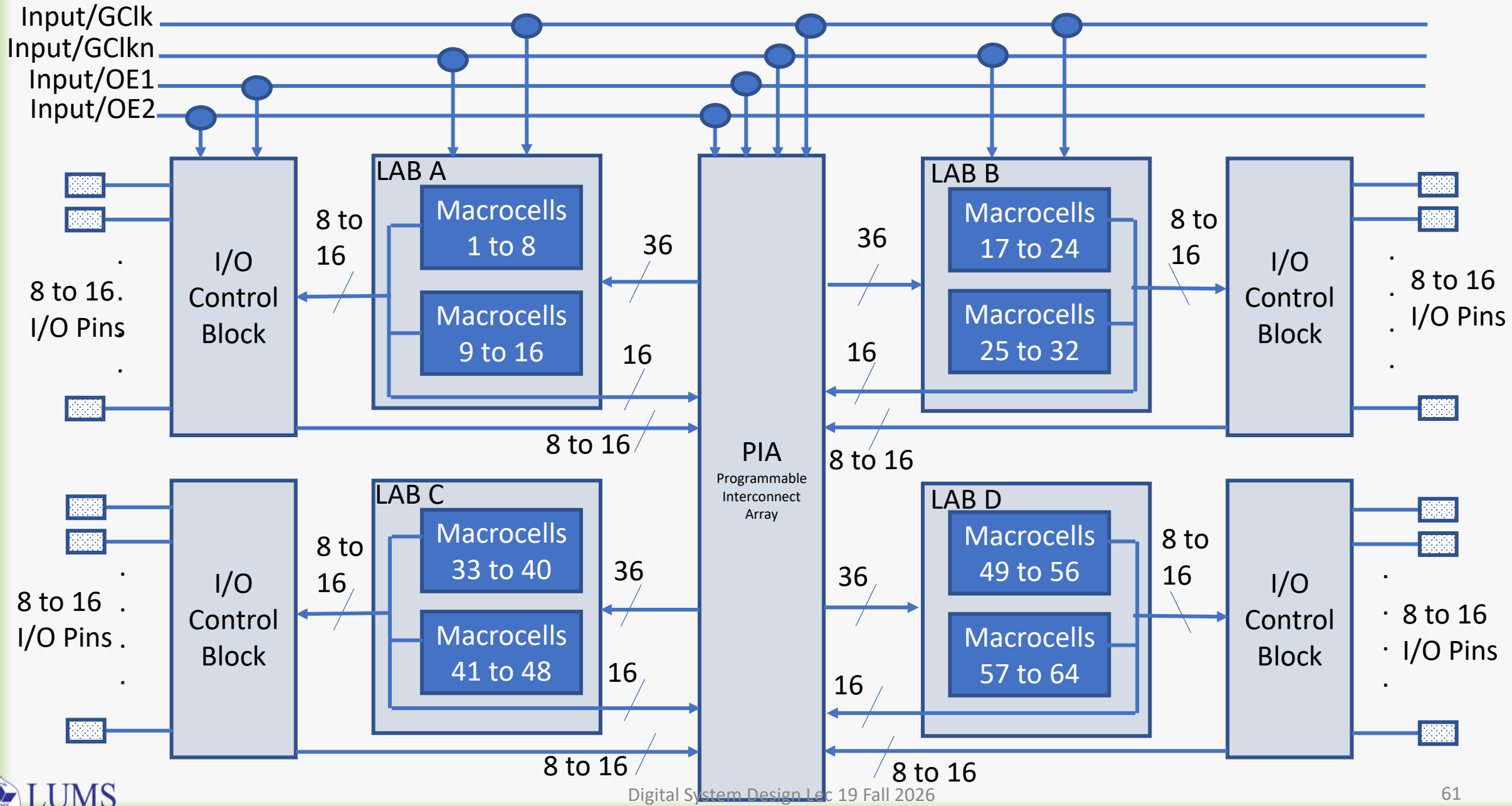
Description of CPLD

- Structured Array of PLD blocks
- Programmable On-Chip Interconnection Fabric
- Have higher number of inputs without much increase in area
- 100% connectivity between macrocells, BUT not all outputs of macrocells are connected to OUTPUT
- Wide Fan-In AND gates are used at INPUTS
- Outputs of macrocells can be routed within the chip to form more complex logic blocks
- The Interconnect Array selects signals from I/O blocks or macrocell outputs and connects them back to function block inputs
- The I/O blocks provide an interface between bidirectional physical I/O pins and the interior of the CPLD

Details of Generic CPLD Configuration

- Collection of individual PLD gate arrays on a single integrated circuit
- Each I/O pin is driven by a 3-state buffer and can be programmed to act as Input or Output pin
- Each PLD typically contains 8 to 16 macrocells
- If a macrocell has unused product terms, they can be used by nearby macrocells
- The outputs of macrocells can be routed to inputs of other macrocells to form complex multi-level logic functions
- Different manufactures offer variations in PLD design, programmable switching matrix and different types of I/O blocks

Altera 7000 Series CPLD Architecture



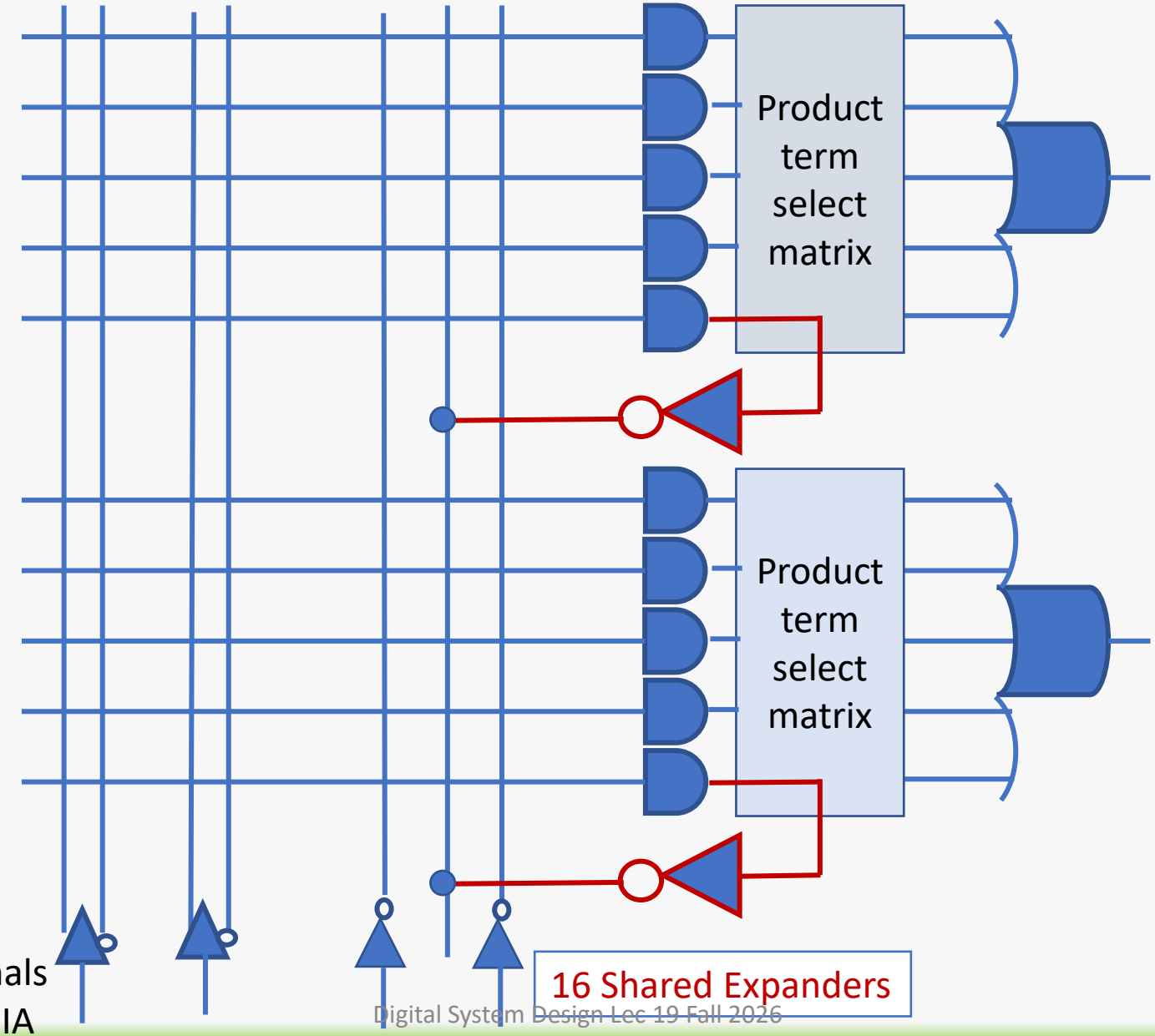
Details of Altera 7000 CPLD

- Floating gate EEPROM Configuration memory cells
- Architecture consists of an array of logic array blocks (**LAB**), programmable interconnect array (**PIA**) and an array of programmable I/O blocks
- Each LAB has 36 inputs and 16 outputs
- Each LAB consists of 16 macrocells providing both combinational and sequential logic
- PIA provides full connectivity between LAB and I/O Cells
- I/O Control block connects I/O Pins and PIA and LAB
- Global input clk (GCLK) and Global Clear (GCLR) connect to all macrocells
- Each macrocell can provide five product terms to its product-term-select matrix

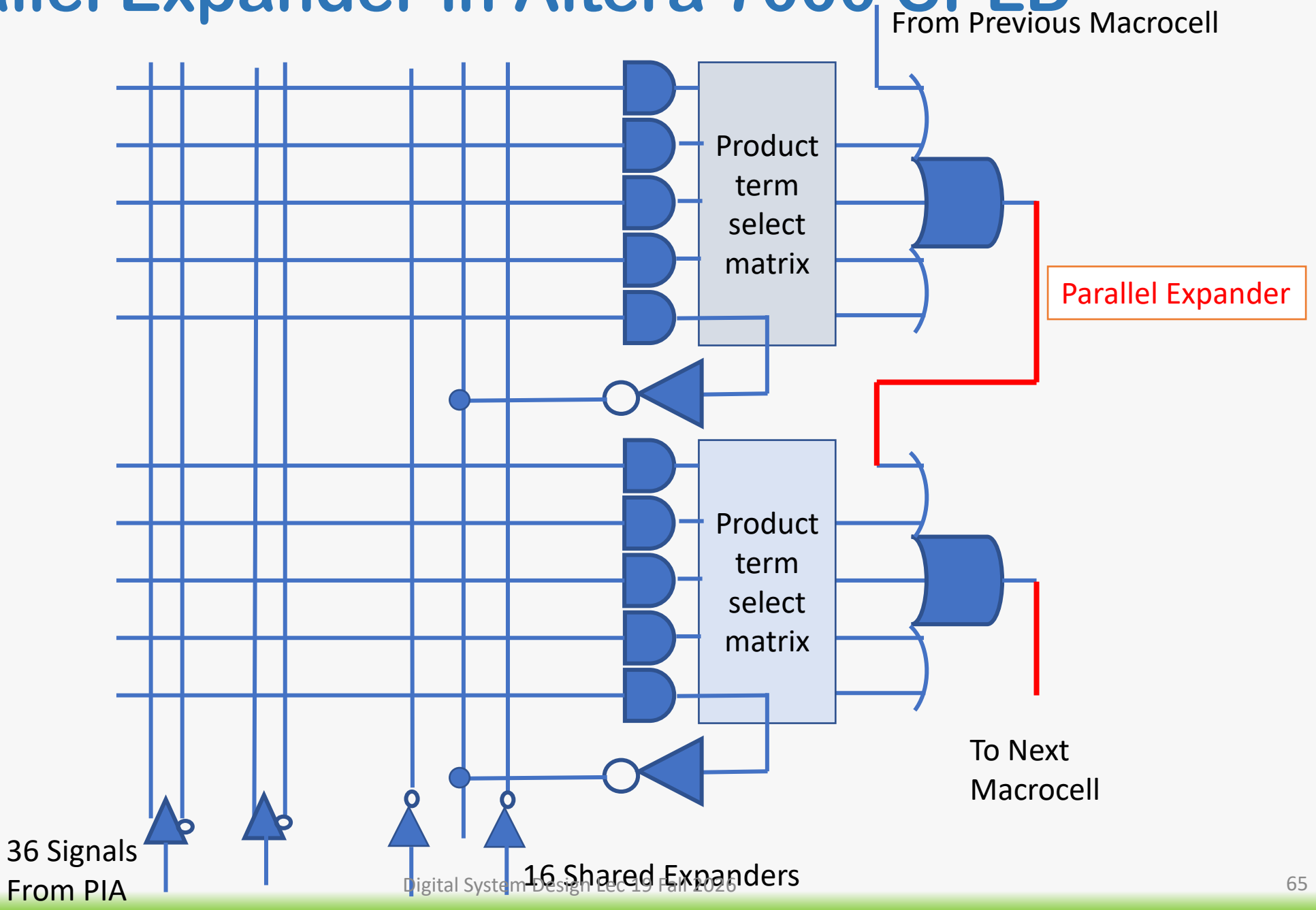
PIA and I/O

- The PIA provides full connectivity with predictable timing between LAB, dedicated inputs and I/O Pins
- The I/O Control block establishes connectivity between I/O Pins and PIA and LABs
- From 8 to 16 LAB outputs can be programmed to route through I/O Pins; and from 8 to 16 I/O Pins can be programmed to route through the I/O Control block to the PIA
- Altera architecture uses a PIA to connect LABs and I/O Pins. All signals are available throughout the device. The timing is thus predictable. The alternative is 'Channel-Routed Architecture' that is more denser but the timing is un-predictable as delays are routing-dependent

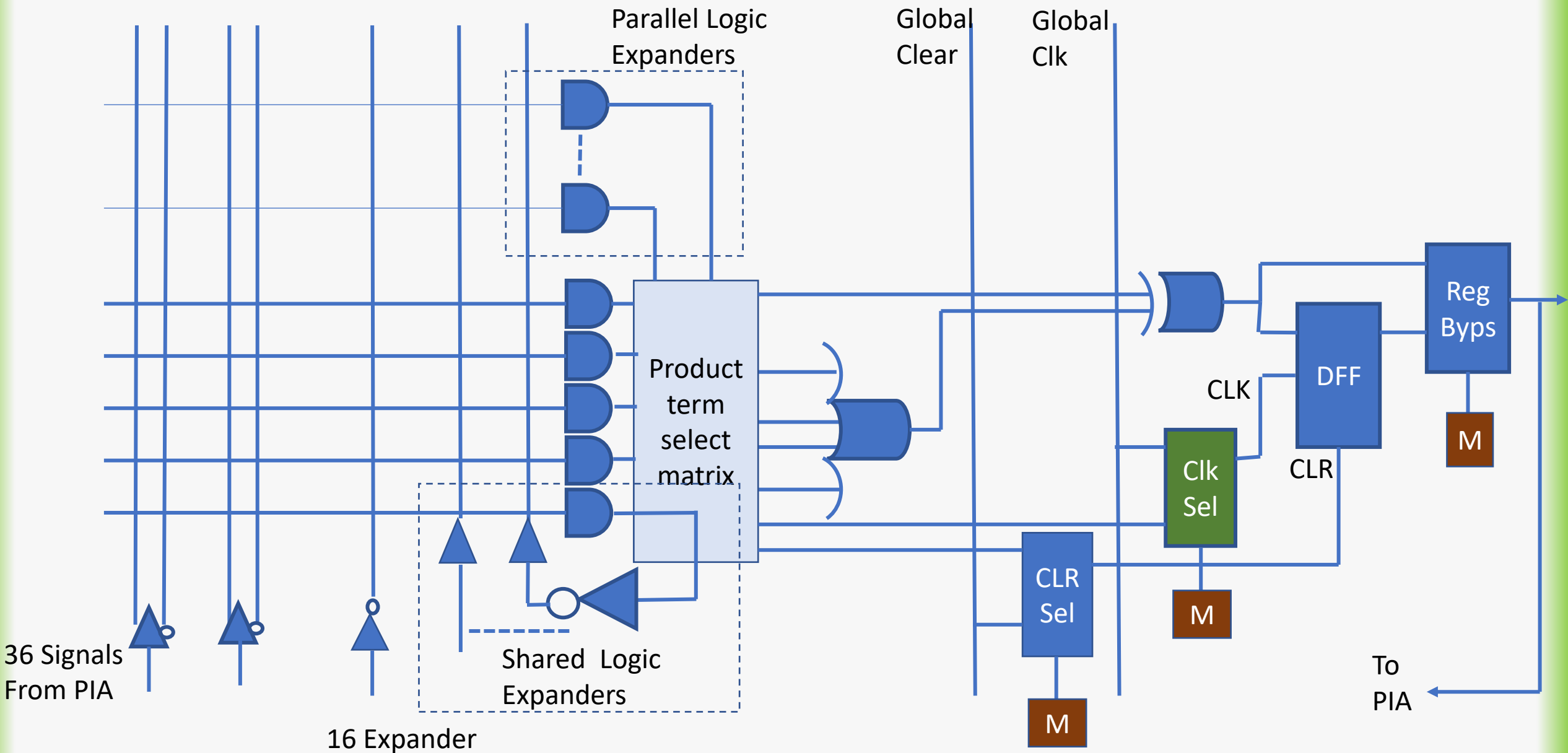
Shareable Expander in Altera 7000 CPLD



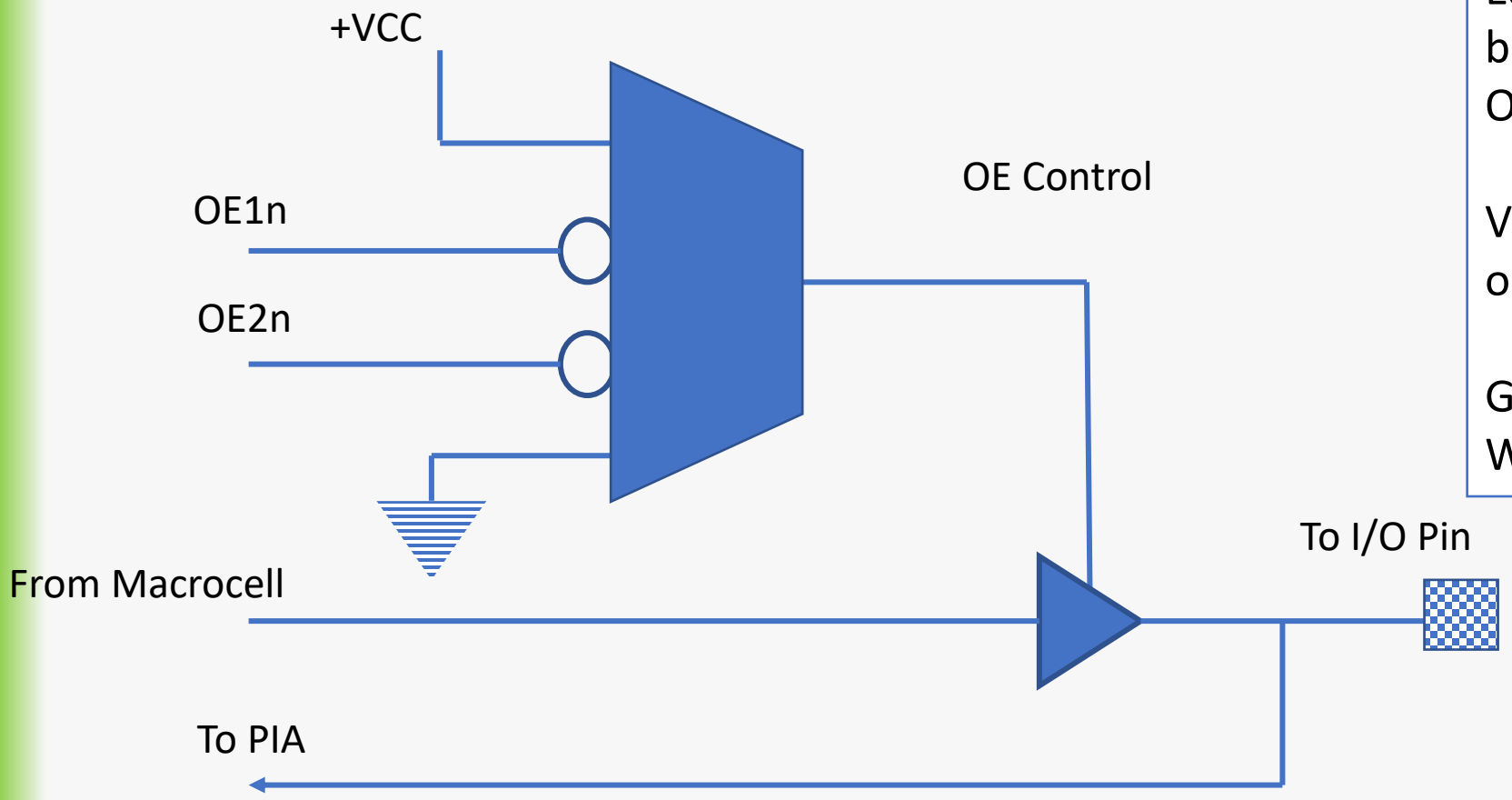
Parallel Expander in Altera 7000 CPLD



Altera 7000 Series Macrocell Architecture



Altera 7000 Series I/O Control Block

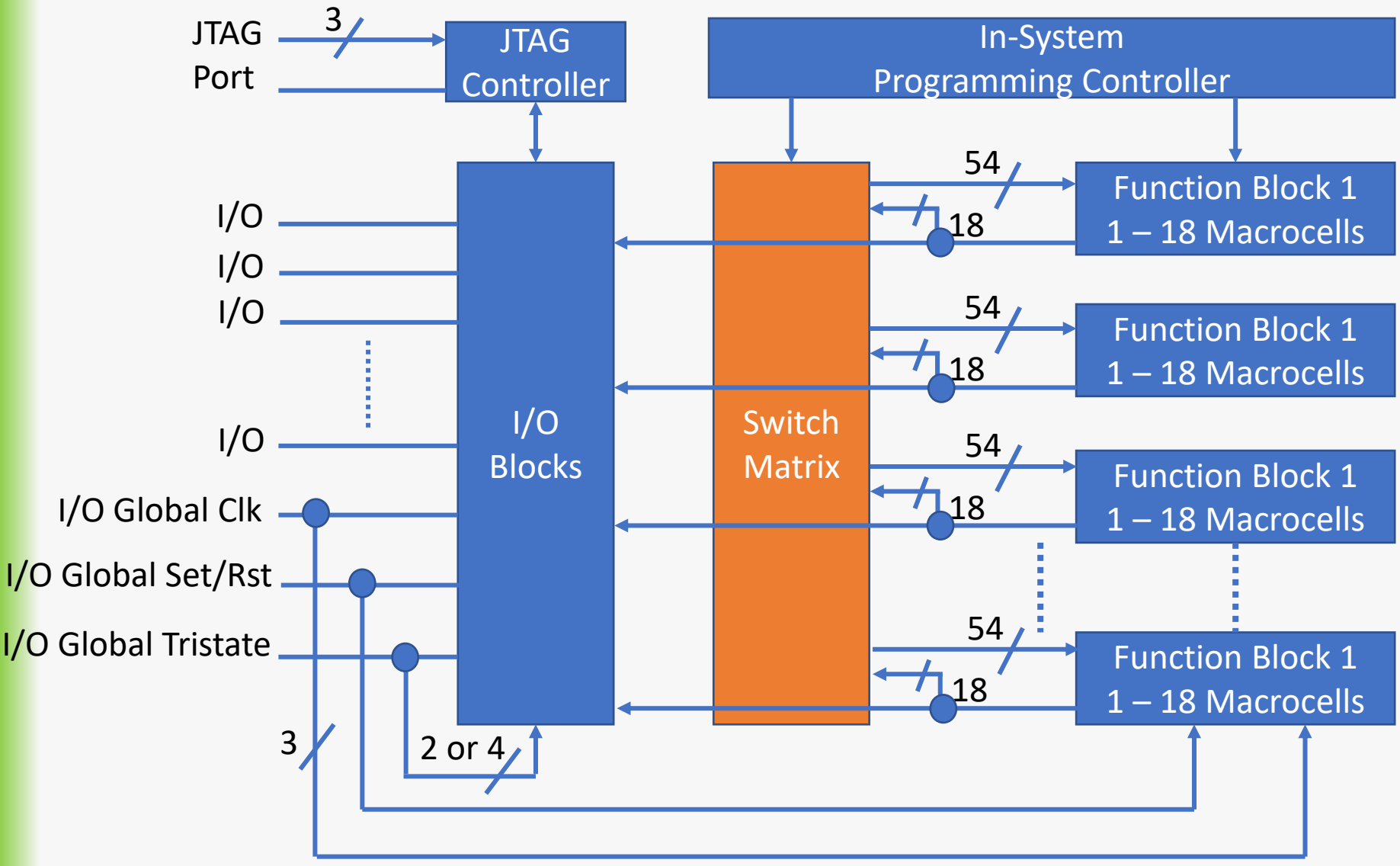


Each I/O Pin can be configured to be dedicated input pin, output pin Or bi-dir under control of OE1, OE2.

VCC will hardwire the I/O pin to output of macrocell.

GND will isolate macrocell and I/O pin Will be an input pin only.

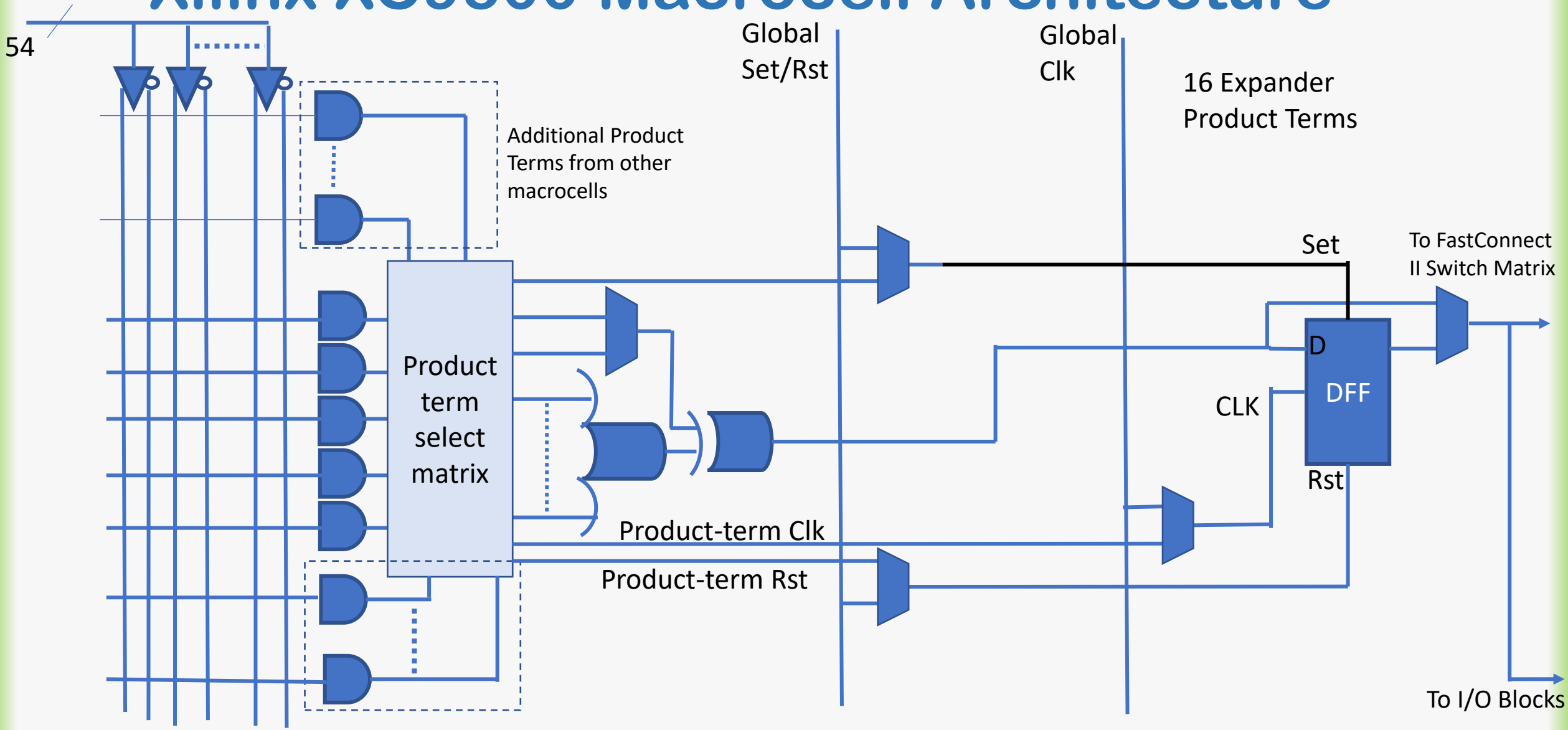
Xilinx XC9500 CPLD Architecture



Details of Xilinx 9500 CPLD

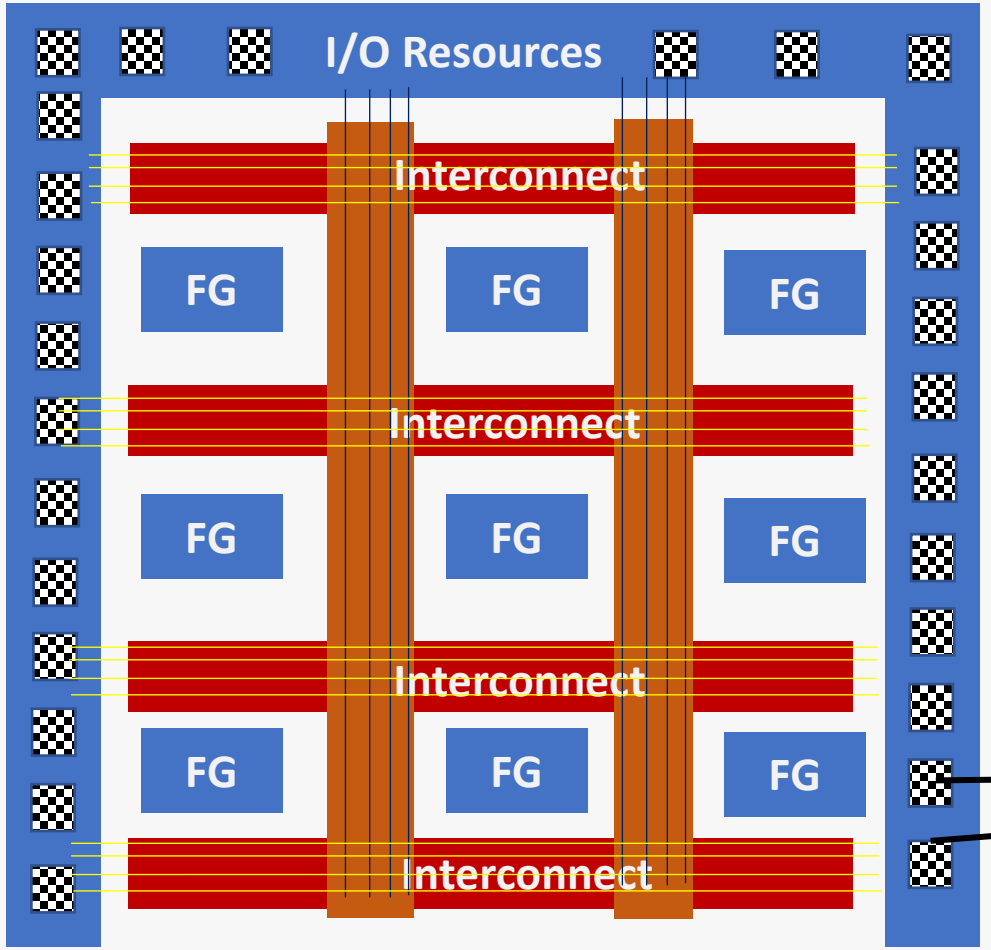
- Flash based EEPROM Technology
- Comprises Few (4) Functional Blocks
- Each functional block contains 18 independent macrocells
- Each functional block has 54 inputs and 18 outputs
- Product-term-allocator allocates upto 90 product terms in each macrocell in a functional block to form SOP expressions
- Each macrocell can receive five direct product terms from the AND array and up to 10 more product terms can be made available from other un-committed product terms in other macrocells from same functional block
- Partial SOP terms can be combined from several macrocells
- Each macrocell can be independently configured for combinational or sequential functionality

Xilinx XC9500 Macrocell Architecture



Field Programmable Gate Arrays
FPGA

Field Programmable Gate Arrays Generic Architecture



FG = function generator for Logic implementation

I/O Blocks



USB or JTAG Cable

SRAM Configuration Memory

Lookup Table (LUT) based Configurable Logic Blocks

